Musical examples

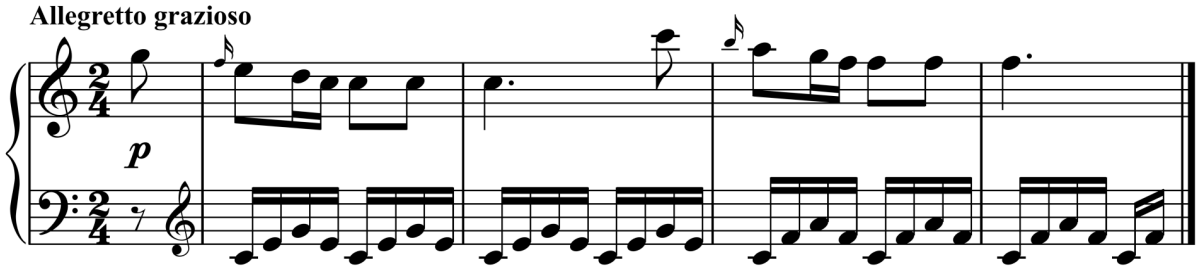
The musical examples, including scores and audios, links to the official Analyzing Classical Form website.

Among the https://acf.research.mcgill.ca/supp_list.phpcomplete list of example we look at how the form inference model perform on only a few of them.

Model output

Identifying presentation

example 2-11



🔊 **Click to Play**

We distinguish between bindings for idea and bindings for formal unit binding. For the presentation, it is the idea that is repeats, while for the period the repeating thing is the formal unit (ABAC)

```
Input surface:
size = 11
  {a ⟦I None⟧}
  {a ⟦IV None⟧}



Output analysis:
  size=11 compressionRate=1.0
  Seq
    idea = "a"
    parameters = ⟦I None⟧
                 ⟦IV None⟧



  size=11 compressionRate=1.0
  Concatenate
    {"a" ⟦I None⟧}
    {"a" ⟦IV None⟧}
expanded analysis:
  SeqF (Label "\"a\"") [Param [I] None,Param [IV] None]
    UseIdeaWithF (Label "\"a\"") (Param [I] None)
      LeafF (Label "\"a\"") (Param [I] None)

    UseIdeaWithF (Label "\"a\"") (Param [IV] None)
      LeafF (Label "\"a\"") (Param [IV] None)
```

```
  ConcatenateF [] []
    LeafF (Label "\"a\"") (Param [I] None)
      LeafF (Label "\"a\"") (Param [I] None)

    LeafF (Label "\"a\"") (Param [IV] None)
      LeafF (Label "\"a\"") (Param [IV] None)
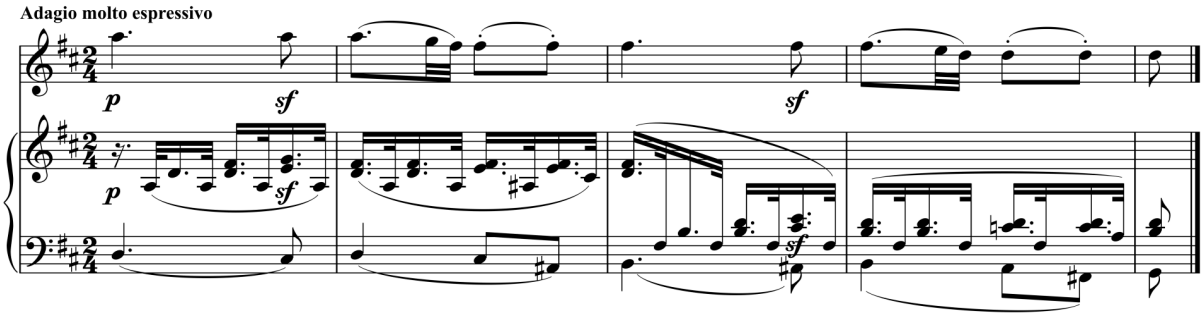
re-generated surface
  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"a\"") (Param [IV] None)



  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"a\"") (Param [IV] None)
```

This example illustrates the basics of how variable binding leads to smaller size. By default, we assign a high cost of raw repetition markers from the input (as they are placeholders for the surface motivic ideas). On the other hand, calling a binding variable just cost 1.

example 2-12



🔊 **Click to Play**

This time we make the input a little bit harder in that the basic idea is not an atomic marker, but the concatenation of two. And the model need to identify the repetition of “a”+“b” as the basic idea, and correctly build up the phrase parameter for the concatenated phrases.

```
Input surface:
size = 27
  {a ⟦I-V None⟧}
  {b ⟦I-III None⟧}
  {a ⟦VI-II None⟧}
  {b ⟦VI-III None⟧}



Output analysis:
  size=25 compressionRate=1.08
  Seq
    idea = "a"+"b"
    parameters = ConcatParam
                   ⟦I-V None⟧
                   ⟦I-III None⟧
                 ConcatParam
                   ⟦VI-II None⟧
                   ⟦VI-III None⟧



  size=27 compressionRate=1.0
  Concatenate
    {"a" ⟦I-V None⟧}
    Concatenate
      {"b" ⟦I-III None⟧}
      Concatenate
        {"a" ⟦VI-II None⟧}
        {"b" ⟦VI-III None⟧}
expanded analysis:
  SeqF (Concat (Label "\"a\"") (Label "\"b\"")) [ConcatParam (Param [I,V] None)
(Param [I,III] None),ConcatParam (Param [VI,II] None) (Param [VI,III] None)]
    UseIdeaWithF (Concat (Label "\"a\"") (Label "\"b\"")) (ConcatParam (Param [I,V]
None) (Param [I,III] None))
      LeafF (Label "\"a\"") (Param [I,V] None)

      LeafF (Label "\"b\"") (Param [I,III] None)
```

```
    UseIdeaWithF (Concat (Label "\"a\"") (Label "\"b\"")) (ConcatParam (Param [VI,II]
None) (Param [VI,III] None))
      LeafF (Label "\"a\"") (Param [VI,II] None)

      LeafF (Label "\"b\"") (Param [VI,III] None)




  ConcatenateF [] []
    LeafF (Label "\"a\"") (Param [I,V] None)
      LeafF (Label "\"a\"") (Param [I,V] None)

    ConcatenateF [] []
      LeafF (Label "\"b\"") (Param [I,III] None)
        LeafF (Label "\"b\"") (Param [I,III] None)

      ConcatenateF [] []
        LeafF (Label "\"a\"") (Param [VI,II] None)
          LeafF (Label "\"a\"") (Param [VI,II] None)

        LeafF (Label "\"b\"") (Param [VI,III] None)
          LeafF (Label "\"b\"") (Param [VI,III] None)
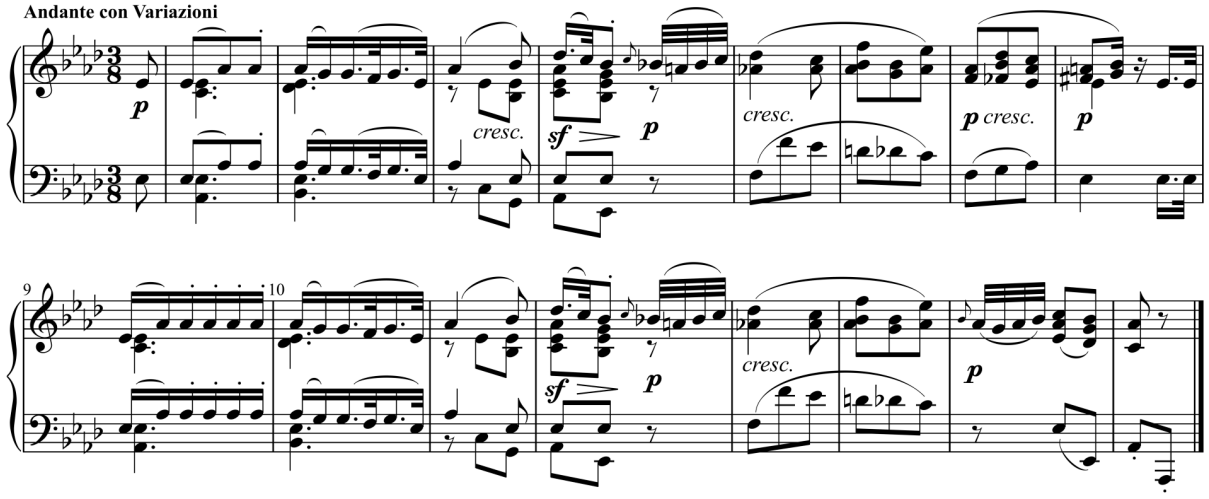
re-generated surface
  LeafF (Label "\"a\"") (Param [I,V] None)
  LeafF (Label "\"b\"") (Param [I,III] None)
  LeafF (Label "\"a\"") (Param [VI,II] None)
  LeafF (Label "\"b\"") (Param [VI,III] None)



  LeafF (Label "\"a\"") (Param [I,V] None)
  LeafF (Label "\"b\"") (Param [I,III] None)
  LeafF (Label "\"a\"") (Param [VI,II] None)
  LeafF (Label "\"b\"") (Param [VI,III] None)
```

Identifying compound themes

example 6-1



🔊 Click to Play

```
Input surface:
size = 60
  {a ⟦I-V None⟧}
  {b ⟦I-V-I-V Weak⟧}
  {c ⟦IV-V-I None⟧}
  {d ⟦VI-VII-I-V Weak⟧}
  {a ⟦I-V None⟧}
  {b ⟦I-V-I-V Weak⟧}
  {c ⟦V None⟧}
  {e ⟦I Strong⟧}



Output analysis:
  size=59 compressionRate=1.0169491525423728
  Period
    antecedentHead = "a"+"b"
    antecedent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ⟦I-V None⟧ ⟦I-V-I-V Weak⟧
                   Concatenate
                     {"c" ⟦IV-V-I None⟧}
                     {"d" ⟦VI-VII-I-V Weak⟧}
    consequent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ⟦I-V None⟧ ⟦I-V-I-V Weak⟧
                   Concatenate
                     {"c" ⟦V None⟧}
                     {"e" ⟦I Strong⟧}



  size=61 compressionRate=0.9836065573770492
  Concatenate
    {"a" ⟦I-V None⟧}
    Concatenate
```

```
      {"b" ⟦I-V-I-V Weak⟧}
      Period
        antecedentHead = "c"
        antecedent = Concatenate
                       UseIdeaWith   $0 ⟦IV-V-I None⟧
                       Concatenate
                         {"d" ⟦VI-VII-I-V Weak⟧}
                         Concatenate
                           {"a" ⟦I-V None⟧}
                           {"b" ⟦I-V-I-V Weak⟧}
        consequent = Concatenate
                       UseIdeaWith   $0 ⟦V None⟧
                       {"e" ⟦I Strong⟧}
expanded analysis:
  PeriodF {basicIdeaF = Concat (Label "\"a\"") (Label "\"b\""), antecedentF = [],
consequentF = []}
    ConcatenateF [Concat (Label "\"a\"") (Label "\"b\"")] [Concat (Label "\"a\"")
(Label "\"b\"")]
      UseIdeaWithF (LocalVar 0) (ConcatParam (Param [I,V] None) (Param [I,V,I,V]
Weak))
        LeafF (Label "\"a\"") (Param [I,V] None)

        LeafF (Label "\"b\"") (Param [I,V,I,V] Weak)

      ConcatenateF [Concat (Label "\"a\"") (Label "\"b\"")] [Concat (Label "\"a\"")
(Label "\"b\"")]
        LeafF (Label "\"c\"") (Param [IV,V,I] None)
          LeafF (Label "\"c\"") (Param [IV,V,I] None)

        LeafF (Label "\"d\"") (Param [VI,VII,I,V] Weak)
          LeafF (Label "\"d\"") (Param [VI,VII,I,V] Weak)

    ConcatenateF [Concat (Label "\"a\"") (Label "\"b\"")] [Concat (Label "\"a\"")
(Label "\"b\"")]
      UseIdeaWithF (LocalVar 0) (ConcatParam (Param [I,V] None) (Param [I,V,I,V]
Weak))
        LeafF (Label "\"a\"") (Param [I,V] None)

        LeafF (Label "\"b\"") (Param [I,V,I,V] Weak)

      ConcatenateF [Concat (Label "\"a\"") (Label "\"b\"")] [Concat (Label "\"a\"")
(Label "\"b\"")]
        LeafF (Label "\"c\"") (Param [V] None)
          LeafF (Label "\"c\"") (Param [V] None)

        LeafF (Label "\"e\"") (Param [I] Strong)
          LeafF (Label "\"e\"") (Param [I] Strong)




  ConcatenateF [] []
    LeafF (Label "\"a\"") (Param [I,V] None)
      LeafF (Label "\"a\"") (Param [I,V] None)

    ConcatenateF [] []
```

```
      LeafF (Label "\"b\"") (Param [I,V,I,V] Weak)
        LeafF (Label "\"b\"") (Param [I,V,I,V] Weak)

      PeriodF {basicIdeaF = Label "\"c\"", antecedentF = [], consequentF = []}
        ConcatenateF [Label "\"c\""] [Label "\"c\""]
          UseIdeaWithF (LocalVar 0) (Param [IV,V,I] None)
            LeafF (Label "\"c\"") (Param [IV,V,I] None)

          ConcatenateF [Label "\"c\""] [Label "\"c\""]
            LeafF (Label "\"d\"") (Param [VI,VII,I,V] Weak)
              LeafF (Label "\"d\"") (Param [VI,VII,I,V] Weak)

            ConcatenateF [Label "\"c\""] [Label "\"c\""]
              LeafF (Label "\"a\"") (Param [I,V] None)
                LeafF (Label "\"a\"") (Param [I,V] None)

              LeafF (Label "\"b\"") (Param [I,V,I,V] Weak)
                LeafF (Label "\"b\"") (Param [I,V,I,V] Weak)

        ConcatenateF [Label "\"c\""] [Label "\"c\""]
          UseIdeaWithF (LocalVar 0) (Param [V] None)
            LeafF (Label "\"c\"") (Param [V] None)
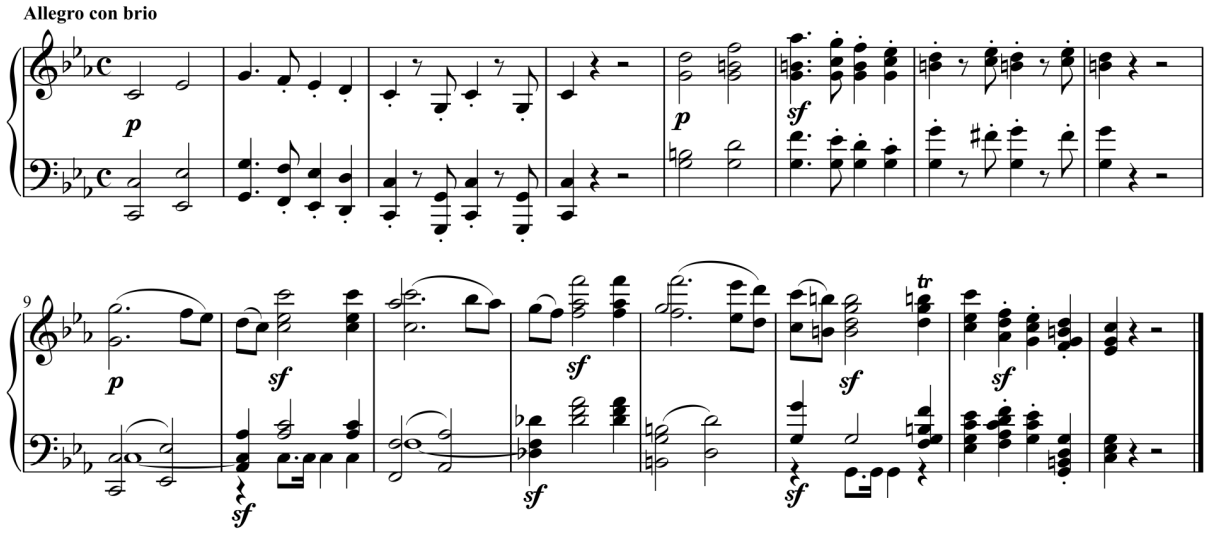
          LeafF (Label "\"e\"") (Param [I] Strong)
            LeafF (Label "\"e\"") (Param [I] Strong)

re-generated surface
  LeafF (Label "\"a\"") (Param [I,V] None)
  LeafF (Label "\"b\"") (Param [I,V,I,V] Weak)
  LeafF (Label "\"c\"") (Param [IV,V,I] None)
  LeafF (Label "\"d\"") (Param [VI,VII,I,V] Weak)
  LeafF (Label "\"a\"") (Param [I,V] None)
  LeafF (Label "\"b\"") (Param [I,V,I,V] Weak)
  LeafF (Label "\"c\"") (Param [V] None)
  LeafF (Label "\"e\"") (Param [I] Strong)



  LeafF (Label "\"a\"") (Param [I,V] None)
  LeafF (Label "\"b\"") (Param [I,V,I,V] Weak)
  LeafF (Label "\"c\"") (Param [IV,V,I] None)
  LeafF (Label "\"d\"") (Param [VI,VII,I,V] Weak)
  LeafF (Label "\"a\"") (Param [I,V] None)
  LeafF (Label "\"b\"") (Param [I,V,I,V] Weak)
  LeafF (Label "\"c\"") (Param [V] None)
  LeafF (Label "\"e\"") (Param [I] Strong)
```

example 6-2



🔊 **Click to Play**

```
Input surface:
size = 64
  {a ⟦I None⟧}
  {b ⟦I None⟧}
  {c ⟦I None⟧}
  {a ⟦V None⟧}
  {b ⟦V None⟧}
  {c ⟦V None⟧}
  {b ⟦I-VI None⟧}
  {b ⟦IV-II None⟧}
  {b ⟦V None⟧}
  {d ⟦I-II-V-I Strong⟧}



Output analysis:
  size=61 compressionRate=1.0491803278688525
  Concatenate
    Seq
      idea = "a"+"b"+"c"
      parameters = ConcatParam
                     ⟦I None⟧
                     ConcatParam
                       ⟦I None⟧
                       ⟦I None⟧
                   ConcatParam
                     ⟦V None⟧
                     ConcatParam
                       ⟦V None⟧
                       ⟦V None⟧
    Sentence
      basicIdea = "b"
      presentation = Seq
                       idea = $0
                       parameters = ⟦I-VI None⟧
```

```
                                    ⟦IV-II None⟧
      continuation = Concatenate
                       {$0 ⟦V None⟧}
                       {"d" ⟦I-II-V-I Strong⟧}



  size=60 compressionRate=1.0666666666666667
  Concatenate
    Concatenate
      Seq
        idea = "a"+"b"+"c"
        parameters = ConcatParam
                       ⟦I None⟧
                       ConcatParam
                         ⟦I None⟧
                         ⟦I None⟧
                     ConcatParam
                       ⟦V None⟧
                       ConcatParam
                         ⟦V None⟧
                         ⟦V None⟧
      {"b" ⟦I-VI None⟧}
    Concatenate
      Seq
        idea = "b"
        parameters = ⟦IV-II None⟧
                     ⟦V None⟧
      {"d" ⟦I-II-V-I Strong⟧}
expanded analysis:
  ConcatenateF [] []
    SeqF (Concat (Label "\"a\"") (Concat (Label "\"b\"") (Label "\"c\"")))
[ConcatParam (Param [I] None) (ConcatParam (Param [I] None) (Param [I]
None)),ConcatParam (Param [V] None) (ConcatParam (Param [V] None) (Param [V] None))]
      UseIdeaWithF (Concat (Label "\"a\"") (Concat (Label "\"b\"") (Label "\"c\"")))
(ConcatParam (Param [I] None) (ConcatParam (Param [I] None) (Param [I] None)))
        LeafF (Label "\"a\"") (Param [I] None)

        LeafF (Label "\"b\"") (Param [I] None)

        LeafF (Label "\"c\"") (Param [I] None)

      UseIdeaWithF (Concat (Label "\"a\"") (Concat (Label "\"b\"") (Label "\"c\"")))
(ConcatParam (Param [V] None) (ConcatParam (Param [V] None) (Param [V] None)))
        LeafF (Label "\"a\"") (Param [V] None)

        LeafF (Label "\"b\"") (Param [V] None)

        LeafF (Label "\"c\"") (Param [V] None)

    SentenceF {basicIdeaF = Label "\"b\"", presentationF = [], continuationF = []}
      SeqF (LocalVar 0) [Param [I,VI] None,Param [IV,II] None]
        UseIdeaWithF (LocalVar 0) (Param [I,VI] None)
          LeafF (Label "\"b\"") (Param [I,VI] None)

        UseIdeaWithF (LocalVar 0) (Param [IV,II] None)
```

```
          LeafF (Label "\"b\"") (Param [IV,II] None)

      ConcatenateF [Label "\"b\""] [Label "\"b\""]
        LeafF (LocalVar 0) (Param [V] None)
          LeafF (Label "\"b\"") (Param [V] None)

        LeafF (Label "\"d\"") (Param [I,II,V,I] Strong)
          LeafF (Label "\"d\"") (Param [I,II,V,I] Strong)




  ConcatenateF [] []
    ConcatenateF [] []
      SeqF (Concat (Label "\"a\"") (Concat (Label "\"b\"") (Label "\"c\""))) [ConcatParam (Param [I] None) (ConcatParam (Param [I] None) (Param [I] None)),ConcatParam (Param [V] None) (ConcatParam (Param [V] None) (Param [V] None))]
        UseIdeaWithF (Concat (Label "\"a\"") (Concat (Label "\"b\"") (Label "\"c\""))) (ConcatParam (Param [I] None) (ConcatParam (Param [I] None) (Param [I] None)))
          LeafF (Label "\"a\"") (Param [I] None)

          LeafF (Label "\"b\"") (Param [I] None)

          LeafF (Label "\"c\"") (Param [I] None)

        UseIdeaWithF (Concat (Label "\"a\"") (Concat (Label "\"b\"") (Label "\"c\""))) (ConcatParam (Param [V] None) (ConcatParam (Param [V] None) (Param [V] None)))
          LeafF (Label "\"a\"") (Param [V] None)

          LeafF (Label "\"b\"") (Param [V] None)

          LeafF (Label "\"c\"") (Param [V] None)

      LeafF (Label "\"b\"") (Param [I,VI] None)
        LeafF (Label "\"b\"") (Param [I,VI] None)

    ConcatenateF [] []
      SeqF (Label "\"b\"") [Param [IV,II] None,Param [V] None]
        UseIdeaWithF (Label "\"b\"") (Param [IV,II] None)
          LeafF (Label "\"b\"") (Param [IV,II] None)

        UseIdeaWithF (Label "\"b\"") (Param [V] None)
          LeafF (Label "\"b\"") (Param [V] None)

      LeafF (Label "\"d\"") (Param [I,II,V,I] Strong)
        LeafF (Label "\"d\"") (Param [I,II,V,I] Strong)
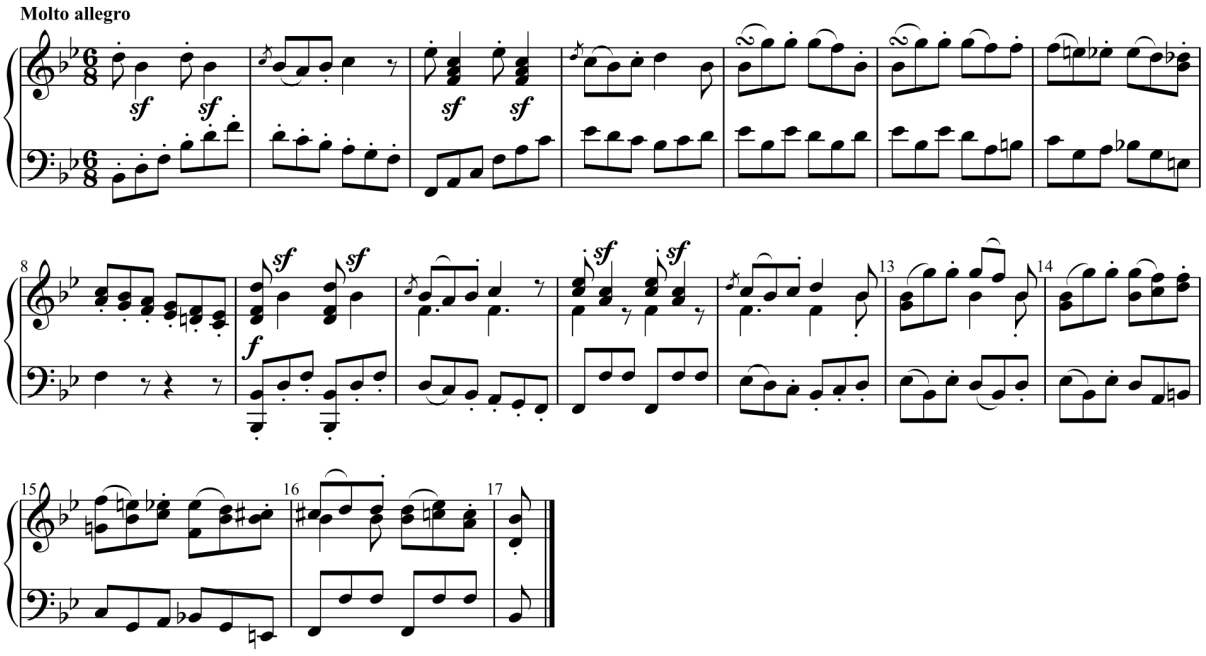
re-generated surface
  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"b\"") (Param [I] None)
  LeafF (Label "\"c\"") (Param [I] None)
  LeafF (Label "\"a\"") (Param [V] None)
  LeafF (Label "\"b\"") (Param [V] None)
  LeafF (Label "\"c\"") (Param [V] None)
```

```
LeafF (Label "\"b\"") (Param [I,VI] None)
LeafF (Label "\"b\"") (Param [IV,II] None)
LeafF (Label "\"b\"") (Param [V] None)
LeafF (Label "\"d\"") (Param [I,II,V,I] Strong)



LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"b\"") (Param [I] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"a\"") (Param [V] None)
LeafF (Label "\"b\"") (Param [V] None)
LeafF (Label "\"c\"") (Param [V] None)
LeafF (Label "\"b\"") (Param [I,VI] None)
LeafF (Label "\"b\"") (Param [IV,II] None)
LeafF (Label "\"b\"") (Param [V] None)
LeafF (Label "\"d\"") (Param [I,II,V,I] Strong)
```

example 6-5



🔊 Click to Play

```
Input surface:
size = 92
  {a ⟦I-V None⟧}
  {a ⟦V-I None⟧}
  {b ⟦IV-I None⟧}
  {b ⟦IV-I None⟧}
  {c ⟦II None⟧}
  {c ⟦I None⟧}
  {d ⟦V Weak⟧}
  {a ⟦I-V None⟧}
  {a ⟦V-I None⟧}
  {b ⟦IV-I None⟧}
  {b ⟦IV-I None⟧}
  {c ⟦II None⟧}
  {c ⟦I None⟧}
  {e ⟦V-I Strong⟧}



Output analysis:
  size=80 compressionRate=1.15
  Period
    antecedentHead = Twice "a"+Twice "b"+Twice "c"
    antecedent = Concatenate
                   UseIdeaWith
                     $0
                     ConcatParam
                       ConcatParam
                         ⟦I-V None⟧
                         ConcatParam
                           ⟦V-I None⟧
```

```
                        ⟦IV-I None⟧
                    ConcatParam
                      ⟦IV-I None⟧
                      ConcatParam
                        ⟦II None⟧
                        ⟦I None⟧
                {"d" ⟦V Weak⟧}
  consequent = Concatenate
                UseIdeaWith
                  $0
                  ConcatParam
                    ConcatParam
                      ⟦I-V None⟧
                      ConcatParam
                        ⟦V-I None⟧
                        ⟦IV-I None⟧
                    ConcatParam
                      ⟦IV-I None⟧
                      ConcatParam
                        ⟦II None⟧
                        ⟦I None⟧
                {"e" ⟦V-I Strong⟧}



size=85 compressionRate=1.0823529411764705
Period
  antecedentHead = Twice "a"+Twice "b"
  antecedent = Concatenate
                UseIdeaWith
                  $0
                  ConcatParam
                    ConcatParam
                      ⟦I-V None⟧
                      ⟦V-I None⟧
                    ConcatParam
                      ⟦IV-I None⟧
                      ⟦IV-I None⟧
                Concatenate
                  Seq
                    idea = "c"
                    parameters = ⟦II None⟧
                                 ⟦I None⟧
                  {"d" ⟦V Weak⟧}
  consequent = Concatenate
                UseIdeaWith
                  $0
                  ConcatParam
                    ConcatParam
                      ⟦I-V None⟧
                      ⟦V-I None⟧
                    ConcatParam
                      ⟦IV-I None⟧
                      ⟦IV-I None⟧
                Concatenate
                  Seq
```

```
                      idea = "c"
                      parameters = ⟦II None⟧
                                   ⟦I None⟧
                    {"e" ⟦V-I Strong⟧}
expanded analysis:
  PeriodF {basicIdeaF = Concat (Twice (Label "\"a\"")) (Concat (Twice (Label
"\"b\"")) (Twice (Label "\"c\""))), antecedentF = [], consequentF = []}
    ConcatenateF [Concat (Twice (Label "\"a\"")) (Concat (Twice (Label "\"b\""))
(Twice (Label "\"c\"")))] [Concat (Twice (Label "\"a\"")) (Concat (Twice (Label
"\"b\"")) (Twice (Label "\"c\"")))]
      UseIdeaWithF (LocalVar 0) (ConcatParam (ConcatParam (Param [I,V] None)
(ConcatParam (Param [V,I] None) (Param [IV,I] None))) (ConcatParam (Param [IV,I]
None) (ConcatParam (Param [II] None) (Param [I] None))))
        LeafF (Label "\"a\"") (Param [I,V] None)

        LeafF (Label "\"a\"") (Param [V,I] None)

        LeafF (Label "\"b\"") (Param [IV,I] None)

        LeafF (Label "\"b\"") (Param [IV,I] None)

        LeafF (Label "\"c\"") (Param [II] None)

        LeafF (Label "\"c\"") (Param [I] None)

      LeafF (Label "\"d\"") (Param [V] Weak)
        LeafF (Label "\"d\"") (Param [V] Weak)

    ConcatenateF [Concat (Twice (Label "\"a\"")) (Concat (Twice (Label "\"b\""))
(Twice (Label "\"c\"")))] [Concat (Twice (Label "\"a\"")) (Concat (Twice (Label
"\"b\"")) (Twice (Label "\"c\"")))]
      UseIdeaWithF (LocalVar 0) (ConcatParam (ConcatParam (Param [I,V] None)
(ConcatParam (Param [V,I] None) (Param [IV,I] None))) (ConcatParam (Param [IV,I]
None) (ConcatParam (Param [II] None) (Param [I] None))))
        LeafF (Label "\"a\"") (Param [I,V] None)

        LeafF (Label "\"a\"") (Param [V,I] None)

        LeafF (Label "\"b\"") (Param [IV,I] None)

        LeafF (Label "\"b\"") (Param [IV,I] None)

        LeafF (Label "\"c\"") (Param [II] None)

        LeafF (Label "\"c\"") (Param [I] None)

      LeafF (Label "\"e\"") (Param [V,I] Strong)
        LeafF (Label "\"e\"") (Param [V,I] Strong)




  PeriodF {basicIdeaF = Concat (Twice (Label "\"a\"")) (Twice (Label "\"b\"")),
antecedentF = [], consequentF = []}
    ConcatenateF [Concat (Twice (Label "\"a\"")) (Twice (Label "\"b\""))] [Concat
(Twice (Label "\"a\"")) (Twice (Label "\"b\""))]
```

```
      UseIdeaWithF (LocalVar 0) (ConcatParam (ConcatParam (Param [I,V] None) (Param
[V,I] None)) (ConcatParam (Param [IV,I] None) (Param [IV,I] None)))
        LeafF (Label "\"a\"") (Param [I,V] None)

        LeafF (Label "\"a\"") (Param [V,I] None)

        LeafF (Label "\"b\"") (Param [IV,I] None)

        LeafF (Label "\"b\"") (Param [IV,I] None)

      ConcatenateF [Concat (Twice (Label "\"a\"")) (Twice (Label "\"b\""))] [Concat
(Twice (Label "\"a\"")) (Twice (Label "\"b\""))]
        SeqF (Label "\"c\"") [Param [II] None,Param [I] None]
          UseIdeaWithF (Label "\"c\"") (Param [II] None)
            LeafF (Label "\"c\"") (Param [II] None)

          UseIdeaWithF (Label "\"c\"") (Param [I] None)
            LeafF (Label "\"c\"") (Param [I] None)

        LeafF (Label "\"d\"") (Param [V] Weak)
          LeafF (Label "\"d\"") (Param [V] Weak)

    ConcatenateF [Concat (Twice (Label "\"a\"")) (Twice (Label "\"b\""))] [Concat
(Twice (Label "\"a\"")) (Twice (Label "\"b\""))]
      UseIdeaWithF (LocalVar 0) (ConcatParam (ConcatParam (Param [I,V] None) (Param
[V,I] None)) (ConcatParam (Param [IV,I] None) (Param [IV,I] None)))
        LeafF (Label "\"a\"") (Param [I,V] None)

        LeafF (Label "\"a\"") (Param [V,I] None)

        LeafF (Label "\"b\"") (Param [IV,I] None)

        LeafF (Label "\"b\"") (Param [IV,I] None)

      ConcatenateF [Concat (Twice (Label "\"a\"")) (Twice (Label "\"b\""))] [Concat
(Twice (Label "\"a\"")) (Twice (Label "\"b\""))]
        SeqF (Label "\"c\"") [Param [II] None,Param [I] None]
          UseIdeaWithF (Label "\"c\"") (Param [II] None)
            LeafF (Label "\"c\"") (Param [II] None)

          UseIdeaWithF (Label "\"c\"") (Param [I] None)
            LeafF (Label "\"c\"") (Param [I] None)

        LeafF (Label "\"e\"") (Param [V,I] Strong)
          LeafF (Label "\"e\"") (Param [V,I] Strong)

re-generated surface
  LeafF (Label "\"a\"") (Param [I,V] None)
  LeafF (Label "\"a\"") (Param [V,I] None)
  LeafF (Label "\"b\"") (Param [IV,I] None)
  LeafF (Label "\"b\"") (Param [IV,I] None)
  LeafF (Label "\"c\"") (Param [II] None)
  LeafF (Label "\"c\"") (Param [I] None)
  LeafF (Label "\"d\"") (Param [V] Weak)
  LeafF (Label "\"a\"") (Param [I,V] None)
  LeafF (Label "\"a\"") (Param [V,I] None)
```

```
LeafF (Label "\"b\"") (Param [IV,I] None)
LeafF (Label "\"b\"") (Param [IV,I] None)
LeafF (Label "\"c\"") (Param [II] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"e\"") (Param [V,I] Strong)
```

```
LeafF (Label "\"a\"") (Param [I,V] None)
LeafF (Label "\"a\"") (Param [V,I] None)
LeafF (Label "\"b\"") (Param [IV,I] None)
LeafF (Label "\"b\"") (Param [IV,I] None)
LeafF (Label "\"c\"") (Param [II] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"d\"") (Param [V] Weak)
LeafF (Label "\"a\"") (Param [I,V] None)
LeafF (Label "\"a\"") (Param [V,I] None)
LeafF (Label "\"b\"") (Param [IV,I] None)
LeafF (Label "\"b\"") (Param [IV,I] None)
LeafF (Label "\"c\"") (Param [II] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"e\"") (Param [V,I] Strong)
```

examples from annotation_pilot

```
Input surface:
size = 335
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {C ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {E ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {C ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {E ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}
  {H ⟦I None⟧}
  {I ⟦I None⟧}
  {J ⟦I None⟧}
  {B ⟦I Weak⟧}
  {K ⟦I None⟧}
  {K ⟦I None⟧}
  {L ⟦I None⟧}
  {M ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {N ⟦I None⟧}
  {G' ⟦I Strong⟧}
  {H' ⟦I None⟧}
  {H' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}
  {H ⟦I None⟧}
  {I ⟦I None⟧}
  {J ⟦I None⟧}
  {B ⟦I Weak⟧}
  {K ⟦I None⟧}
  {K ⟦I None⟧}
  {L ⟦I None⟧}
  {M ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {N ⟦I None⟧}
  {G' ⟦I Strong⟧}
  {H' ⟦I None⟧}
```

```
  {H' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}



Output analysis:
  size=146 compressionRate=2.2945205479452055
  SimpleBinary
    aPart = Period
              antecedentHead = "A"+"B"
              antecedent = Concatenate
                             UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                             Concatenate
                               {"C" ⟦I None⟧}
                               {"D" ⟦I Weak⟧}
              consequent = Concatenate
                             UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                             HybridAntCont
                               antecedent = Concatenate
                                              {"E" ⟦I None⟧}
                                              {"D" ⟦I Weak⟧}
                               continuation = Concatenate
                                                TwiceFormalUnit
                                                  {"A'" ⟦I None⟧}
                                                Concatenate
                                                  {"F" ⟦I None⟧}
                                                  {"G" ⟦I Strong⟧}
    bPart = Concatenate
              {"H" ⟦I None⟧}
              Concatenate
                {"I" ⟦I None⟧}
                Concatenate
                  {"J" ⟦I None⟧}
                  Concatenate
                    Concatenate
                      {"B" ⟦I Weak⟧}
                      Concatenate
                        TwiceFormalUnit
                          {"K" ⟦I None⟧}
                        Concatenate
                          {"L" ⟦I None⟧}
                          {"M" ⟦I Weak⟧}
                    Concatenate
                      TwiceFormalUnit
                        {"A'" ⟦I None⟧}
                      Concatenate
                        Concatenate
                          {"N" ⟦I None⟧}
                          {"G'" ⟦I Strong⟧}
                        Concatenate
                          TwiceFormalUnit
                            {"H'" ⟦I None⟧}
                          Concatenate
                            {"F" ⟦I None⟧}
```

```
                            {"G" ⟦I Strong⟧}


  size=148 compressionRate=2.2635135135135136
  Concatenate
    TwiceFormalUnit
      Period
        antecedentHead = "A"+"B"
        antecedent = Concatenate
                       UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                       Concatenate
                         {"C" ⟦I None⟧}
                         {"D" ⟦I Weak⟧}
        consequent = Concatenate
                       UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                       HybridAntCont
                         antecedent = Concatenate
                                        {"E" ⟦I None⟧}
                                        {"D" ⟦I Weak⟧}
                         continuation = Concatenate
                                          TwiceFormalUnit
                                            {"A'" ⟦I None⟧}
                                          Concatenate
                                            {"F" ⟦I None⟧}
                                            {"G" ⟦I Strong⟧}
    TwiceFormalUnit
      Concatenate
        {"H" ⟦I None⟧}
        Concatenate
          {"I" ⟦I None⟧}
          Concatenate
            {"J" ⟦I None⟧}
            Concatenate
              Concatenate
                {"B" ⟦I Weak⟧}
                Concatenate
                  TwiceFormalUnit
                    {"K" ⟦I None⟧}
                  Concatenate
                    {"L" ⟦I None⟧}
                    {"M" ⟦I Weak⟧}
              Concatenate
                TwiceFormalUnit
                  {"A'" ⟦I None⟧}
                Concatenate
                  Concatenate
                    {"N" ⟦I None⟧}
                    {"G'" ⟦I Strong⟧}
                  Concatenate
                    TwiceFormalUnit
                      {"H'" ⟦I None⟧}
                    Concatenate
                      {"F" ⟦I None⟧}
                      {"G" ⟦I Strong⟧}
expanded analysis:
```

```
  SimpleBinaryF [] []
    PeriodF {basicIdeaF = Concat (Label "\"A\"") (Label "\"B\""), antecedentF = [],
consequentF = []}
      ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label "\"A\"")
(Label "\"B\"")]
        UseIdeaWithF (LocalVar 0) (ConcatParam (Param [I] None) (Param [I] None))
          LeafF (Label "\"A\"") (Param [I] None)

          LeafF (Label "\"B\"") (Param [I] None)

        ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label "\"A\"")
(Label "\"B\"")]
          LeafF (Label "\"C\"") (Param [I] None)
            LeafF (Label "\"C\"") (Param [I] None)

          LeafF (Label "\"D\"") (Param [I] Weak)
            LeafF (Label "\"D\"") (Param [I] Weak)

      ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label "\"A\"")
(Label "\"B\"")]
        UseIdeaWithF (LocalVar 0) (ConcatParam (Param [I] None) (Param [I] None))
          LeafF (Label "\"A\"") (Param [I] None)

          LeafF (Label "\"B\"") (Param [I] None)

        HybridAntContF {antecedentF = [Concat (Label "\"A\"") (Label "\"B\"")],
continuationF = [Concat (Label "\"A\"") (Label "\"B\"")]}
          ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label
"\"A\"") (Label "\"B\"")]
            LeafF (Label "\"E\"") (Param [I] None)
              LeafF (Label "\"E\"") (Param [I] None)

            LeafF (Label "\"D\"") (Param [I] Weak)
              LeafF (Label "\"D\"") (Param [I] Weak)

          ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label
"\"A\"") (Label "\"B\"")]
            TwiceFormalUnitF [Concat (Label "\"A\"") (Label "\"B\"")]
              LeafF (Label "\"A'\"") (Param [I] None)
                LeafF (Label "\"A'\"") (Param [I] None)

              LeafF (Label "\"A'\"") (Param [I] None)
                LeafF (Label "\"A'\"") (Param [I] None)

            ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label
"\"A\"") (Label "\"B\"")]
              LeafF (Label "\"F\"") (Param [I] None)
                LeafF (Label "\"F\"") (Param [I] None)

              LeafF (Label "\"G\"") (Param [I] Strong)
                LeafF (Label "\"G\"") (Param [I] Strong)

    PeriodF {basicIdeaF = Concat (Label "\"A\"") (Label "\"B\""), antecedentF = [],
consequentF = []}
      ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label "\"A\"")
(Label "\"B\"")]
```

```
        UseIdeaWithF (LocalVar 0) (ConcatParam (Param [I] None) (Param [I] None))
          LeafF (Label "\"A\"") (Param [I] None)

          LeafF (Label "\"B\"") (Param [I] None)

        ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label "\"A\"")
(Label "\"B\"")]
          LeafF (Label "\"C\"") (Param [I] None)
            LeafF (Label "\"C\"") (Param [I] None)

          LeafF (Label "\"D\"") (Param [I] Weak)
            LeafF (Label "\"D\"") (Param [I] Weak)

      ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label "\"A\"")
(Label "\"B\"")]
        UseIdeaWithF (LocalVar 0) (ConcatParam (Param [I] None) (Param [I] None))
          LeafF (Label "\"A\"") (Param [I] None)

          LeafF (Label "\"B\"") (Param [I] None)

        HybridAntContF {antecedentF = [Concat (Label "\"A\"") (Label "\"B\"")],
continuationF = [Concat (Label "\"A\"") (Label "\"B\"")]}
          ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label
"\"A\"") (Label "\"B\"")]
            LeafF (Label "\"E\"") (Param [I] None)
              LeafF (Label "\"E\"") (Param [I] None)

            LeafF (Label "\"D\"") (Param [I] Weak)
              LeafF (Label "\"D\"") (Param [I] Weak)

          ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label
"\"A\"") (Label "\"B\"")]
            TwiceFormalUnitF [Concat (Label "\"A\"") (Label "\"B\"")]
              LeafF (Label "\"A'\"") (Param [I] None)
                LeafF (Label "\"A'\"") (Param [I] None)

              LeafF (Label "\"A'\"") (Param [I] None)
                LeafF (Label "\"A'\"") (Param [I] None)

            ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label
"\"A\"") (Label "\"B\"")]
              LeafF (Label "\"F\"") (Param [I] None)
                LeafF (Label "\"F\"") (Param [I] None)

              LeafF (Label "\"G\"") (Param [I] Strong)
                LeafF (Label "\"G\"") (Param [I] Strong)

    ConcatenateF [] []
      LeafF (Label "\"H\"") (Param [I] None)
        LeafF (Label "\"H\"") (Param [I] None)

      ConcatenateF [] []
        LeafF (Label "\"I\"") (Param [I] None)
          LeafF (Label "\"I\"") (Param [I] None)

        ConcatenateF [] []
```

```
      LeafF (Label "\"J\"") (Param [I] None)
        LeafF (Label "\"J\"") (Param [I] None)

      ConcatenateF [] []
        ConcatenateF [] []
          LeafF (Label "\"B\"") (Param [I] Weak)
            LeafF (Label "\"B\"") (Param [I] Weak)

          ConcatenateF [] []
            TwiceFormalUnitF []
              LeafF (Label "\"K\"") (Param [I] None)
                LeafF (Label "\"K\"") (Param [I] None)

              LeafF (Label "\"K\"") (Param [I] None)
                LeafF (Label "\"K\"") (Param [I] None)

            ConcatenateF [] []
              LeafF (Label "\"L\"") (Param [I] None)
                LeafF (Label "\"L\"") (Param [I] None)

              LeafF (Label "\"M\"") (Param [I] Weak)
                LeafF (Label "\"M\"") (Param [I] Weak)

        ConcatenateF [] []
          TwiceFormalUnitF []
            LeafF (Label "\"A'\"") (Param [I] None)
              LeafF (Label "\"A'\"") (Param [I] None)

            LeafF (Label "\"A'\"") (Param [I] None)
              LeafF (Label "\"A'\"") (Param [I] None)

          ConcatenateF [] []
            ConcatenateF [] []
              LeafF (Label "\"N\"") (Param [I] None)
                LeafF (Label "\"N\"") (Param [I] None)

              LeafF (Label "\"G'\"") (Param [I] Strong)
                LeafF (Label "\"G'\"") (Param [I] Strong)

            ConcatenateF [] []
              TwiceFormalUnitF []
                LeafF (Label "\"H'\"") (Param [I] None)
                  LeafF (Label "\"H'\"") (Param [I] None)

                LeafF (Label "\"H'\"") (Param [I] None)
                  LeafF (Label "\"H'\"") (Param [I] None)

              ConcatenateF [] []
                LeafF (Label "\"F\"") (Param [I] None)
                  LeafF (Label "\"F\"") (Param [I] None)

                LeafF (Label "\"G\"") (Param [I] Strong)
                  LeafF (Label "\"G\"") (Param [I] Strong)

ConcatenateF [] []
  LeafF (Label "\"H\"") (Param [I] None)
```

```
  LeafF (Label "\"H\"") (Param [I] None)

ConcatenateF [] []
  LeafF (Label "\"I\"") (Param [I] None)
    LeafF (Label "\"I\"") (Param [I] None)

  ConcatenateF [] []
    LeafF (Label "\"J\"") (Param [I] None)
      LeafF (Label "\"J\"") (Param [I] None)

    ConcatenateF [] []
      ConcatenateF [] []
        LeafF (Label "\"B\"") (Param [I] Weak)
          LeafF (Label "\"B\"") (Param [I] Weak)

        ConcatenateF [] []
          TwiceFormalUnitF []
            LeafF (Label "\"K\"") (Param [I] None)
              LeafF (Label "\"K\"") (Param [I] None)

            LeafF (Label "\"K\"") (Param [I] None)
              LeafF (Label "\"K\"") (Param [I] None)

          ConcatenateF [] []
            LeafF (Label "\"L\"") (Param [I] None)
              LeafF (Label "\"L\"") (Param [I] None)

            LeafF (Label "\"M\"") (Param [I] Weak)
              LeafF (Label "\"M\"") (Param [I] Weak)

      ConcatenateF [] []
        TwiceFormalUnitF []
          LeafF (Label "\"A'\"") (Param [I] None)
            LeafF (Label "\"A'\"") (Param [I] None)

          LeafF (Label "\"A'\"") (Param [I] None)
            LeafF (Label "\"A'\"") (Param [I] None)

        ConcatenateF [] []
          ConcatenateF [] []
            LeafF (Label "\"N\"") (Param [I] None)
              LeafF (Label "\"N\"") (Param [I] None)

            LeafF (Label "\"G'\"") (Param [I] Strong)
              LeafF (Label "\"G'\"") (Param [I] Strong)

          ConcatenateF [] []
            TwiceFormalUnitF []
              LeafF (Label "\"H'\"") (Param [I] None)
                LeafF (Label "\"H'\"") (Param [I] None)

              LeafF (Label "\"H'\"") (Param [I] None)
                LeafF (Label "\"H'\"") (Param [I] None)

            ConcatenateF [] []
              LeafF (Label "\"F\"") (Param [I] None)
```

```
                    LeafF (Label "\"F\"") (Param [I] None)

                  LeafF (Label "\"G\"") (Param [I] Strong)
                    LeafF (Label "\"G\"") (Param [I] Strong)



  ConcatenateF [] []
    TwiceFormalUnitF []
      PeriodF {basicIdeaF = Concat (Label "\"A\"") (Label "\"B\""), antecedentF = [],
consequentF = []}
        ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label "\"A\"")
(Label "\"B\"")]
          UseIdeaWithF (LocalVar 0) (ConcatParam (Param [I] None) (Param [I] None))
            LeafF (Label "\"A\"") (Param [I] None)

            LeafF (Label "\"B\"") (Param [I] None)

          ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label
"\"A\"") (Label "\"B\"")]
            LeafF (Label "\"C\"") (Param [I] None)
              LeafF (Label "\"C\"") (Param [I] None)

            LeafF (Label "\"D\"") (Param [I] Weak)
              LeafF (Label "\"D\"") (Param [I] Weak)

        ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label "\"A\"")
(Label "\"B\"")]
          UseIdeaWithF (LocalVar 0) (ConcatParam (Param [I] None) (Param [I] None))
            LeafF (Label "\"A\"") (Param [I] None)

            LeafF (Label "\"B\"") (Param [I] None)

          HybridAntContF {antecedentF = [Concat (Label "\"A\"") (Label "\"B\"")],
continuationF = [Concat (Label "\"A\"") (Label "\"B\"")]}
            ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label
"\"A\"") (Label "\"B\"")]
              LeafF (Label "\"E\"") (Param [I] None)
                LeafF (Label "\"E\"") (Param [I] None)

              LeafF (Label "\"D\"") (Param [I] Weak)
                LeafF (Label "\"D\"") (Param [I] Weak)

            ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label
"\"A\"") (Label "\"B\"")]
              TwiceFormalUnitF [Concat (Label "\"A\"") (Label "\"B\"")]
                LeafF (Label "\"A'\"") (Param [I] None)
                  LeafF (Label "\"A'\"") (Param [I] None)

                LeafF (Label "\"A'\"") (Param [I] None)
                  LeafF (Label "\"A'\"") (Param [I] None)

              ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label
"\"A\"") (Label "\"B\"")]
                LeafF (Label "\"F\"") (Param [I] None)
```

```
                LeafF (Label "\"F\"") (Param [I] None)

              LeafF (Label "\"G\"") (Param [I] Strong)
                LeafF (Label "\"G\"") (Param [I] Strong)

      PeriodF {basicIdeaF = Concat (Label "\"A\"") (Label "\"B\""), antecedentF = [],
consequentF = []}
        ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label "\"A\"")
(Label "\"B\"")]
          UseIdeaWithF (LocalVar 0) (ConcatParam (Param [I] None) (Param [I] None))
            LeafF (Label "\"A\"") (Param [I] None)

            LeafF (Label "\"B\"") (Param [I] None)

          ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label
"\"A\"") (Label "\"B\"")]
            LeafF (Label "\"C\"") (Param [I] None)
              LeafF (Label "\"C\"") (Param [I] None)

            LeafF (Label "\"D\"") (Param [I] Weak)
              LeafF (Label "\"D\"") (Param [I] Weak)

        ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label "\"A\"")
(Label "\"B\"")]
          UseIdeaWithF (LocalVar 0) (ConcatParam (Param [I] None) (Param [I] None))
            LeafF (Label "\"A\"") (Param [I] None)

            LeafF (Label "\"B\"") (Param [I] None)

          HybridAntContF {antecedentF = [Concat (Label "\"A\"") (Label "\"B\"")],
continuationF = [Concat (Label "\"A\"") (Label "\"B\"")]}
            ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label
"\"A\"") (Label "\"B\"")]
              LeafF (Label "\"E\"") (Param [I] None)
                LeafF (Label "\"E\"") (Param [I] None)

              LeafF (Label "\"D\"") (Param [I] Weak)
                LeafF (Label "\"D\"") (Param [I] Weak)

            ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label
"\"A\"") (Label "\"B\"")]
              TwiceFormalUnitF [Concat (Label "\"A\"") (Label "\"B\"")]
                LeafF (Label "\"A'\"") (Param [I] None)
                  LeafF (Label "\"A'\"") (Param [I] None)

                LeafF (Label "\"A'\"") (Param [I] None)
                  LeafF (Label "\"A'\"") (Param [I] None)

              ConcatenateF [Concat (Label "\"A\"") (Label "\"B\"")] [Concat (Label
"\"A\"") (Label "\"B\"")]
                LeafF (Label "\"F\"") (Param [I] None)
                  LeafF (Label "\"F\"") (Param [I] None)

                LeafF (Label "\"G\"") (Param [I] Strong)
                  LeafF (Label "\"G\"") (Param [I] Strong)
```

```
TwiceFormalUnitF []
  ConcatenateF [] []
    LeafF (Label "\"H\"") (Param [I] None)
      LeafF (Label "\"H\"") (Param [I] None)

    ConcatenateF [] []
      LeafF (Label "\"I\"") (Param [I] None)
        LeafF (Label "\"I\"") (Param [I] None)

      ConcatenateF [] []
        LeafF (Label "\"J\"") (Param [I] None)
          LeafF (Label "\"J\"") (Param [I] None)

        ConcatenateF [] []
          ConcatenateF [] []
            LeafF (Label "\"B\"") (Param [I] Weak)
              LeafF (Label "\"B\"") (Param [I] Weak)

            ConcatenateF [] []
              TwiceFormalUnitF []
                LeafF (Label "\"K\"") (Param [I] None)
                  LeafF (Label "\"K\"") (Param [I] None)

                LeafF (Label "\"K\"") (Param [I] None)
                  LeafF (Label "\"K\"") (Param [I] None)

              ConcatenateF [] []
                LeafF (Label "\"L\"") (Param [I] None)
                  LeafF (Label "\"L\"") (Param [I] None)

                LeafF (Label "\"M\"") (Param [I] Weak)
                  LeafF (Label "\"M\"") (Param [I] Weak)

          ConcatenateF [] []
            TwiceFormalUnitF []
              LeafF (Label "\"A'\"") (Param [I] None)
                LeafF (Label "\"A'\"") (Param [I] None)

              LeafF (Label "\"A'\"") (Param [I] None)
                LeafF (Label "\"A'\"") (Param [I] None)

            ConcatenateF [] []
              ConcatenateF [] []
                LeafF (Label "\"N\"") (Param [I] None)
                  LeafF (Label "\"N\"") (Param [I] None)

                LeafF (Label "\"G'\"") (Param [I] Strong)
                  LeafF (Label "\"G'\"") (Param [I] Strong)

              ConcatenateF [] []
                TwiceFormalUnitF []
                  LeafF (Label "\"H'\"") (Param [I] None)
                    LeafF (Label "\"H'\"") (Param [I] None)

                  LeafF (Label "\"H'\"") (Param [I] None)
                    LeafF (Label "\"H'\"") (Param [I] None)
```

```
              ConcatenateF [] []
                LeafF (Label "\"F\"") (Param [I] None)
                  LeafF (Label "\"F\"") (Param [I] None)

                LeafF (Label "\"G\"") (Param [I] Strong)
                  LeafF (Label "\"G\"") (Param [I] Strong)

ConcatenateF [] []
  LeafF (Label "\"H\"") (Param [I] None)
    LeafF (Label "\"H\"") (Param [I] None)

  ConcatenateF [] []
    LeafF (Label "\"I\"") (Param [I] None)
      LeafF (Label "\"I\"") (Param [I] None)

    ConcatenateF [] []
      LeafF (Label "\"J\"") (Param [I] None)
        LeafF (Label "\"J\"") (Param [I] None)

      ConcatenateF [] []
        ConcatenateF [] []
          LeafF (Label "\"B\"") (Param [I] Weak)
            LeafF (Label "\"B\"") (Param [I] Weak)

          ConcatenateF [] []
            TwiceFormalUnitF []
              LeafF (Label "\"K\"") (Param [I] None)
                LeafF (Label "\"K\"") (Param [I] None)

              LeafF (Label "\"K\"") (Param [I] None)
                LeafF (Label "\"K\"") (Param [I] None)

            ConcatenateF [] []
              LeafF (Label "\"L\"") (Param [I] None)
                LeafF (Label "\"L\"") (Param [I] None)

              LeafF (Label "\"M\"") (Param [I] Weak)
                LeafF (Label "\"M\"") (Param [I] Weak)

        ConcatenateF [] []
          TwiceFormalUnitF []
            LeafF (Label "\"A'\"") (Param [I] None)
              LeafF (Label "\"A'\"") (Param [I] None)

            LeafF (Label "\"A'\"") (Param [I] None)
              LeafF (Label "\"A'\"") (Param [I] None)

          ConcatenateF [] []
            ConcatenateF [] []
              LeafF (Label "\"N\"") (Param [I] None)
                LeafF (Label "\"N\"") (Param [I] None)

              LeafF (Label "\"G'\"") (Param [I] Strong)
                LeafF (Label "\"G'\"") (Param [I] Strong)
```

```
                    ConcatenateF [] []
                      TwiceFormalUnitF []
                        LeafF (Label "\"H'\"") (Param [I] None)
                          LeafF (Label "\"H'\"") (Param [I] None)

                        LeafF (Label "\"H'\"") (Param [I] None)
                          LeafF (Label "\"H'\"") (Param [I] None)

                      ConcatenateF [] []
                        LeafF (Label "\"F\"") (Param [I] None)
                          LeafF (Label "\"F\"") (Param [I] None)

                        LeafF (Label "\"G\"") (Param [I] Strong)
                          LeafF (Label "\"G\"") (Param [I] Strong)

re-generated surface
  LeafF (Label "\"A\"") (Param [I] None)
  LeafF (Label "\"B\"") (Param [I] None)
  LeafF (Label "\"C\"") (Param [I] None)
  LeafF (Label "\"D\"") (Param [I] Weak)
  LeafF (Label "\"A\"") (Param [I] None)
  LeafF (Label "\"B\"") (Param [I] None)
  LeafF (Label "\"E\"") (Param [I] None)
  LeafF (Label "\"D\"") (Param [I] Weak)
  LeafF (Label "\"A'\"") (Param [I] None)
  LeafF (Label "\"A'\"") (Param [I] None)
  LeafF (Label "\"F\"") (Param [I] None)
  LeafF (Label "\"G\"") (Param [I] Strong)
  LeafF (Label "\"A\"") (Param [I] None)
  LeafF (Label "\"B\"") (Param [I] None)
  LeafF (Label "\"C\"") (Param [I] None)
  LeafF (Label "\"D\"") (Param [I] Weak)
  LeafF (Label "\"A\"") (Param [I] None)
  LeafF (Label "\"B\"") (Param [I] None)
  LeafF (Label "\"E\"") (Param [I] None)
  LeafF (Label "\"D\"") (Param [I] Weak)
  LeafF (Label "\"A'\"") (Param [I] None)
  LeafF (Label "\"A'\"") (Param [I] None)
  LeafF (Label "\"F\"") (Param [I] None)
  LeafF (Label "\"G\"") (Param [I] Strong)
  LeafF (Label "\"H\"") (Param [I] None)
  LeafF (Label "\"I\"") (Param [I] None)
  LeafF (Label "\"J\"") (Param [I] None)
  LeafF (Label "\"B\"") (Param [I] Weak)
  LeafF (Label "\"K\"") (Param [I] None)
  LeafF (Label "\"K\"") (Param [I] None)
  LeafF (Label "\"L\"") (Param [I] None)
  LeafF (Label "\"M\"") (Param [I] Weak)
  LeafF (Label "\"A'\"") (Param [I] None)
  LeafF (Label "\"A'\"") (Param [I] None)
  LeafF (Label "\"N\"") (Param [I] None)
  LeafF (Label "\"G'\"") (Param [I] Strong)
  LeafF (Label "\"H'\"") (Param [I] None)
  LeafF (Label "\"H'\"") (Param [I] None)
  LeafF (Label "\"F\"") (Param [I] None)
  LeafF (Label "\"G\"") (Param [I] Strong)
```

```
LeafF (Label "\"H\"") (Param [I] None)
LeafF (Label "\"I\"") (Param [I] None)
LeafF (Label "\"J\"") (Param [I] None)
LeafF (Label "\"B\"") (Param [I] Weak)
LeafF (Label "\"K\"") (Param [I] None)
LeafF (Label "\"K\"") (Param [I] None)
LeafF (Label "\"L\"") (Param [I] None)
LeafF (Label "\"M\"") (Param [I] Weak)
LeafF (Label "\"A'\"") (Param [I] None)
LeafF (Label "\"A'\"") (Param [I] None)
LeafF (Label "\"N\"") (Param [I] None)
LeafF (Label "\"G'\"") (Param [I] Strong)
LeafF (Label "\"H'\"") (Param [I] None)
LeafF (Label "\"H'\"") (Param [I] None)
LeafF (Label "\"F\"") (Param [I] None)
LeafF (Label "\"G\"") (Param [I] Strong)



LeafF (Label "\"A\"") (Param [I] None)
LeafF (Label "\"B\"") (Param [I] None)
LeafF (Label "\"C\"") (Param [I] None)
LeafF (Label "\"D\"") (Param [I] Weak)
LeafF (Label "\"A\"") (Param [I] None)
LeafF (Label "\"B\"") (Param [I] None)
LeafF (Label "\"E\"") (Param [I] None)
LeafF (Label "\"D\"") (Param [I] Weak)
LeafF (Label "\"A'\"") (Param [I] None)
LeafF (Label "\"A'\"") (Param [I] None)
LeafF (Label "\"F\"") (Param [I] None)
LeafF (Label "\"G\"") (Param [I] Strong)
LeafF (Label "\"A\"") (Param [I] None)
LeafF (Label "\"B\"") (Param [I] None)
LeafF (Label "\"C\"") (Param [I] None)
LeafF (Label "\"D\"") (Param [I] Weak)
LeafF (Label "\"A\"") (Param [I] None)
LeafF (Label "\"B\"") (Param [I] None)
LeafF (Label "\"E\"") (Param [I] None)
LeafF (Label "\"D\"") (Param [I] Weak)
LeafF (Label "\"A'\"") (Param [I] None)
LeafF (Label "\"A'\"") (Param [I] None)
LeafF (Label "\"F\"") (Param [I] None)
LeafF (Label "\"G\"") (Param [I] Strong)
LeafF (Label "\"H\"") (Param [I] None)
LeafF (Label "\"I\"") (Param [I] None)
LeafF (Label "\"J\"") (Param [I] None)
LeafF (Label "\"B\"") (Param [I] Weak)
LeafF (Label "\"K\"") (Param [I] None)
LeafF (Label "\"K\"") (Param [I] None)
LeafF (Label "\"L\"") (Param [I] None)
LeafF (Label "\"M\"") (Param [I] Weak)
LeafF (Label "\"A'\"") (Param [I] None)
LeafF (Label "\"A'\"") (Param [I] None)
LeafF (Label "\"N\"") (Param [I] None)
LeafF (Label "\"G'\"") (Param [I] Strong)
LeafF (Label "\"H'\"") (Param [I] None)
```

```
LeafF (Label "\"H'\"") (Param [I] None)
LeafF (Label "\"F\"") (Param [I] None)
LeafF (Label "\"G\"") (Param [I] Strong)
LeafF (Label "\"H\"") (Param [I] None)
LeafF (Label "\"I\"") (Param [I] None)
LeafF (Label "\"J\"") (Param [I] None)
LeafF (Label "\"B\"") (Param [I] Weak)
LeafF (Label "\"K\"") (Param [I] None)
LeafF (Label "\"K\"") (Param [I] None)
LeafF (Label "\"L\"") (Param [I] None)
LeafF (Label "\"M\"") (Param [I] Weak)
LeafF (Label "\"A'\"") (Param [I] None)
LeafF (Label "\"A'\"") (Param [I] None)
LeafF (Label "\"N\"") (Param [I] None)
LeafF (Label "\"G'\"") (Param [I] Strong)
LeafF (Label "\"H'\"") (Param [I] None)
LeafF (Label "\"H'\"") (Param [I] None)
LeafF (Label "\"F\"") (Param [I] None)
LeafF (Label "\"G\"") (Param [I] Strong)
```

```
Input surface:
size = 623
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}
  {g ⟦I None⟧}
  {h ⟦I None⟧}
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {g ⟦I None⟧}
  {h ⟦I None⟧}
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {g ⟦I None⟧}
  {k ⟦IV None⟧}
  {g ⟦II None⟧}
  {k ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦I None⟧}
```

```
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {g ⟦I None⟧}
  {k ⟦IV None⟧}
  {g ⟦II None⟧}
  {k ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦I None⟧}
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}



Output analysis:
  size=154 compressionRate=4.045454545454546
```

```
Ternary
  exposition = SimpleBinary
                aPart = HybridAntCont
                          antecedent = Sentence
                                         basicIdea = "a"
                                         presentation = Seq
                                                          idea = $0
                                                          parameters = ⟦I None⟧
                                                                       ⟦V None⟧
                                         continuation = Concatenate
                                                          {$0 ⟦I None⟧}
                                                          {"b" ⟦V Weak⟧}
                                continuation = Concatenate
                                                 TwiceFormalUnit
                                                   {"c" ⟦I None⟧}
                                                 Concatenate
                                                   {"d" ⟦I None⟧}
                                                   {"e" ⟦I Strong⟧}
                bPart = HybridAntCont
                          antecedent = Concatenate
                                         Seq
                                           idea = "a"
                                           parameters = ⟦I None⟧
                                                        ⟦V None⟧
                                         Concatenate
                                           {"f" ⟦V None⟧}
                                           {"b" ⟦V Weak⟧}
                                continuation = Concatenate
                                                 TwiceFormalUnit
                                                   {"c" ⟦I None⟧}
                                                 Concatenate
                                                   {"d" ⟦V None⟧}
                                                   {"e" ⟦I Strong⟧}
  contrastingMiddile = RoundedBinary
                         exposition = Concatenate
                                        Seq
                                          idea = "g"+"h"
                                          parameters = ConcatParam
                                                         ⟦I None⟧
                                                         ⟦I None⟧
                                                       ConcatParam
                                                         ⟦V None⟧
                                                         ⟦V None⟧
                                        Concatenate
                                          Concatenate
                                            {"g" ⟦I None⟧}
                                            {"h" ⟦IV None⟧}
                                          Concatenate
                                            {"i" ⟦V None⟧}
                                            {"j" ⟦I Strong⟧}
                         contrastingMiddile = Seq
                                                idea = "g"+"k"
                                                parameters = ConcatParam
                                                               ⟦I None⟧
                                                               ⟦IV None⟧
                                                             ConcatParam
```

```
                                                                  ⟦II None⟧
                                                                  ⟦V None⟧


size=201 compressionRate=3.099502487562189
Concatenate
  TwiceFormalUnit
    HybridAntCont
      antecedent = Sentence
                     basicIdea = "a"
                     presentation = Seq
                                      idea = $0
                                      parameters = ⟦I None⟧
                                                   ⟦V None⟧
                     continuation = Concatenate
                                      {$0 ⟦I None⟧}
                                      {"b" ⟦V Weak⟧}
      continuation = Concatenate
                       TwiceFormalUnit
                         {"c" ⟦I None⟧}
                       Concatenate
                         {"d" ⟦I None⟧}
                         {"e" ⟦I Strong⟧}
  Ternary
    exposition = TwiceFormalUnit
                   HybridAntCont
                     antecedent = Concatenate
                                    Seq
                                      idea = "a"
                                      parameters = ⟦I None⟧
                                                   ⟦V None⟧
                                    Concatenate
                                      {"f" ⟦V None⟧}
                                      {"b" ⟦V Weak⟧}
                     continuation = Concatenate
                                      TwiceFormalUnit
                                        {"c" ⟦I None⟧}
                                      Concatenate
                                        {"d" ⟦V None⟧}
                                        {"e" ⟦I Strong⟧}
    contrastingMiddile = Concatenate
                           RoundedBinary
                             exposition = Concatenate
                                            Seq
                                              idea = "g"+"h"
                                              parameters = ConcatParam
                                                             ⟦I None⟧
                                                             ⟦I None⟧
                                                           ConcatParam
                                                             ⟦V None⟧
                                                             ⟦V None⟧
                                            Concatenate
                                              Concatenate
                                                {"g" ⟦I None⟧}
                                                {"h" ⟦IV None⟧}
```

```
                                                  Concatenate
                                                    {"i" ⟦V None⟧}
                                                    {"j" ⟦I Strong⟧}
                                    contrastingMiddle = Seq
                                                        idea = "g"+"k"
                                                        parameters = ConcatParam
                                                                       ⟦I None⟧
                                                                       ⟦IV None⟧
                                                                     ConcatParam
                                                                       ⟦II None⟧
                                                                       ⟦V None⟧
                                  TwiceFormalUnit
                                    HybridAntCont
                                      antecedent = Sentence
                                                   basicIdea = "a"
                                                   presentation = Seq
                                                                  idea = $0
                                                                  parameters = ⟦I
None⟧
                                                                                ⟦V
None⟧
                                                   continuation = Concatenate
                                                                  {$0 ⟦I None⟧}
                                                                  {"b" ⟦V Weak⟧}
                                      continuation = Concatenate
                                                       TwiceFormalUnit
                                                         {"c" ⟦I None⟧}
                                                       Concatenate
                                                         {"d" ⟦I None⟧}
                                                         {"e" ⟦I Strong⟧}
expanded analysis:
  TernaryF [] []
    SimpleBinaryF [] []
      HybridAntContF {antecedentF = [], continuationF = []}
        SentenceF {basicIdeaF = Label "\"a\"", presentationF = [], continuationF =
[]}
          SeqF (LocalVar 0) [Param [I] None,Param [V] None]
            UseIdeaWithF (LocalVar 0) (Param [I] None)
              LeafF (Label "\"a\"") (Param [I] None)

            UseIdeaWithF (LocalVar 0) (Param [V] None)
              LeafF (Label "\"a\"") (Param [V] None)

          ConcatenateF [Label "\"a\""] [Label "\"a\""]
            LeafF (LocalVar 0) (Param [I] None)
              LeafF (Label "\"a\"") (Param [I] None)

            LeafF (Label "\"b\"") (Param [V] Weak)
              LeafF (Label "\"b\"") (Param [V] Weak)

        ConcatenateF [] []
          TwiceFormalUnitF []
            LeafF (Label "\"c\"") (Param [I] None)
              LeafF (Label "\"c\"") (Param [I] None)

            LeafF (Label "\"c\"") (Param [I] None)
```

```
              LeafF (Label "\"c\"") (Param [I] None)

          ConcatenateF [] []
            LeafF (Label "\"d\"") (Param [I] None)
              LeafF (Label "\"d\"") (Param [I] None)

            LeafF (Label "\"e\"") (Param [I] Strong)
              LeafF (Label "\"e\"") (Param [I] Strong)

      HybridAntContF {antecedentF = [], continuationF = []}
        SentenceF {basicIdeaF = Label "\"a\"", presentationF = [], continuationF =
[]}
          SeqF (LocalVar 0) [Param [I] None,Param [V] None]
            UseIdeaWithF (LocalVar 0) (Param [I] None)
              LeafF (Label "\"a\"") (Param [I] None)

            UseIdeaWithF (LocalVar 0) (Param [V] None)
              LeafF (Label "\"a\"") (Param [V] None)

          ConcatenateF [Label "\"a\""] [Label "\"a\""]
            LeafF (LocalVar 0) (Param [I] None)
              LeafF (Label "\"a\"") (Param [I] None)

            LeafF (Label "\"b\"") (Param [V] Weak)
              LeafF (Label "\"b\"") (Param [V] Weak)

        ConcatenateF [] []
          TwiceFormalUnitF []
            LeafF (Label "\"c\"") (Param [I] None)
              LeafF (Label "\"c\"") (Param [I] None)

            LeafF (Label "\"c\"") (Param [I] None)
              LeafF (Label "\"c\"") (Param [I] None)

          ConcatenateF [] []
            LeafF (Label "\"d\"") (Param [I] None)
              LeafF (Label "\"d\"") (Param [I] None)

            LeafF (Label "\"e\"") (Param [I] Strong)
              LeafF (Label "\"e\"") (Param [I] Strong)

      HybridAntContF {antecedentF = [], continuationF = []}
        ConcatenateF [] []
          SeqF (Label "\"a\"") [Param [I] None,Param [V] None]
            UseIdeaWithF (Label "\"a\"") (Param [I] None)
              LeafF (Label "\"a\"") (Param [I] None)

            UseIdeaWithF (Label "\"a\"") (Param [V] None)
              LeafF (Label "\"a\"") (Param [V] None)

          ConcatenateF [] []
            LeafF (Label "\"f\"") (Param [V] None)
              LeafF (Label "\"f\"") (Param [V] None)

            LeafF (Label "\"b\"") (Param [V] Weak)
              LeafF (Label "\"b\"") (Param [V] Weak)
```

```
        ConcatenateF [] []
          TwiceFormalUnitF []
            LeafF (Label "\"c\"") (Param [I] None)
              LeafF (Label "\"c\"") (Param [I] None)

            LeafF (Label "\"c\"") (Param [I] None)
              LeafF (Label "\"c\"") (Param [I] None)

          ConcatenateF [] []
            LeafF (Label "\"d\"") (Param [V] None)
              LeafF (Label "\"d\"") (Param [V] None)

            LeafF (Label "\"e\"") (Param [I] Strong)
              LeafF (Label "\"e\"") (Param [I] Strong)

      HybridAntContF {antecedentF = [], continuationF = []}
        ConcatenateF [] []
          SeqF (Label "\"a\"") [Param [I] None,Param [V] None]
            UseIdeaWithF (Label "\"a\"") (Param [I] None)
              LeafF (Label "\"a\"") (Param [I] None)

            UseIdeaWithF (Label "\"a\"") (Param [V] None)
              LeafF (Label "\"a\"") (Param [V] None)

          ConcatenateF [] []
            LeafF (Label "\"f\"") (Param [V] None)
              LeafF (Label "\"f\"") (Param [V] None)

            LeafF (Label "\"b\"") (Param [V] Weak)
              LeafF (Label "\"b\"") (Param [V] Weak)

        ConcatenateF [] []
          TwiceFormalUnitF []
            LeafF (Label "\"c\"") (Param [I] None)
              LeafF (Label "\"c\"") (Param [I] None)

            LeafF (Label "\"c\"") (Param [I] None)
              LeafF (Label "\"c\"") (Param [I] None)

          ConcatenateF [] []
            LeafF (Label "\"d\"") (Param [V] None)
              LeafF (Label "\"d\"") (Param [V] None)

            LeafF (Label "\"e\"") (Param [I] Strong)
              LeafF (Label "\"e\"") (Param [I] Strong)

    RoundedBinaryF {expositionF = [], contrastingMiddleF = []}
      ConcatenateF [] []
        SeqF (Concat (Label "\"g\"") (Label "\"h\"")) [ConcatParam (Param [I] None)
(Param [I] None),ConcatParam (Param [V] None) (Param [V] None)]
          UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam (Param
[I] None) (Param [I] None))
            LeafF (Label "\"g\"") (Param [I] None)

            LeafF (Label "\"h\"") (Param [I] None)
```

```
          UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam (Param
[V] None) (Param [V] None))
            LeafF (Label "\"g\"") (Param [V] None)

            LeafF (Label "\"h\"") (Param [V] None)

        ConcatenateF [] []
          ConcatenateF [] []
            LeafF (Label "\"g\"") (Param [I] None)
              LeafF (Label "\"g\"") (Param [I] None)

            LeafF (Label "\"h\"") (Param [IV] None)
              LeafF (Label "\"h\"") (Param [IV] None)

          ConcatenateF [] []
            LeafF (Label "\"i\"") (Param [V] None)
              LeafF (Label "\"i\"") (Param [V] None)

            LeafF (Label "\"j\"") (Param [I] Strong)
              LeafF (Label "\"j\"") (Param [I] Strong)

      ConcatenateF [] []
        SeqF (Concat (Label "\"g\"") (Label "\"h\"")) [ConcatParam (Param [I] None)
(Param [I] None),ConcatParam (Param [V] None) (Param [V] None)]
          UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam (Param
[I] None) (Param [I] None))
            LeafF (Label "\"g\"") (Param [I] None)

            LeafF (Label "\"h\"") (Param [I] None)

          UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam (Param
[V] None) (Param [V] None))
            LeafF (Label "\"g\"") (Param [V] None)

            LeafF (Label "\"h\"") (Param [V] None)

        ConcatenateF [] []
          ConcatenateF [] []
            LeafF (Label "\"g\"") (Param [I] None)
              LeafF (Label "\"g\"") (Param [I] None)

            LeafF (Label "\"h\"") (Param [IV] None)
              LeafF (Label "\"h\"") (Param [IV] None)

          ConcatenateF [] []
            LeafF (Label "\"i\"") (Param [V] None)
              LeafF (Label "\"i\"") (Param [V] None)

            LeafF (Label "\"j\"") (Param [I] Strong)
              LeafF (Label "\"j\"") (Param [I] Strong)

      SeqF (Concat (Label "\"g\"") (Label "\"k\"")) [ConcatParam (Param [I] None)
(Param [IV] None),ConcatParam (Param [II] None) (Param [V] None)]
        UseIdeaWithF (Concat (Label "\"g\"") (Label "\"k\"")) (ConcatParam (Param [I]
None) (Param [IV] None))
```

```
          LeafF (Label "\"g\"") (Param [I] None)

          LeafF (Label "\"k\"") (Param [IV] None)

        UseIdeaWithF (Concat (Label "\"g\"") (Label "\"k\"")) (ConcatParam (Param
[II] None) (Param [V] None))
          LeafF (Label "\"g\"") (Param [II] None)

          LeafF (Label "\"k\"") (Param [V] None)

      ConcatenateF [] []
        SeqF (Concat (Label "\"g\"") (Label "\"h\"")) [ConcatParam (Param [I] None)
(Param [I] None),ConcatParam (Param [V] None) (Param [V] None)]
          UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam (Param
[I] None) (Param [I] None))
            LeafF (Label "\"g\"") (Param [I] None)

            LeafF (Label "\"h\"") (Param [I] None)

          UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam (Param
[V] None) (Param [V] None))
            LeafF (Label "\"g\"") (Param [V] None)

            LeafF (Label "\"h\"") (Param [V] None)

        ConcatenateF [] []
          ConcatenateF [] []
            LeafF (Label "\"g\"") (Param [I] None)
              LeafF (Label "\"g\"") (Param [I] None)

            LeafF (Label "\"h\"") (Param [IV] None)
              LeafF (Label "\"h\"") (Param [IV] None)

          ConcatenateF [] []
            LeafF (Label "\"i\"") (Param [V] None)
              LeafF (Label "\"i\"") (Param [V] None)

            LeafF (Label "\"j\"") (Param [I] Strong)
              LeafF (Label "\"j\"") (Param [I] Strong)

      SeqF (Concat (Label "\"g\"") (Label "\"k\"")) [ConcatParam (Param [I] None)
(Param [IV] None),ConcatParam (Param [II] None) (Param [V] None)]
        UseIdeaWithF (Concat (Label "\"g\"") (Label "\"k\"")) (ConcatParam (Param [I]
None) (Param [IV] None))
          LeafF (Label "\"g\"") (Param [I] None)

          LeafF (Label "\"k\"") (Param [IV] None)

        UseIdeaWithF (Concat (Label "\"g\"") (Label "\"k\"")) (ConcatParam (Param
[II] None) (Param [V] None))
          LeafF (Label "\"g\"") (Param [II] None)

          LeafF (Label "\"k\"") (Param [V] None)

      ConcatenateF [] []
        SeqF (Concat (Label "\"g\"") (Label "\"h\"")) [ConcatParam (Param [I] None)
```

```
(Param [I] None),ConcatParam (Param [V] None) (Param [V] None)]
          UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam (Param
[I] None) (Param [I] None))
            LeafF (Label "\"g\"") (Param [I] None)

            LeafF (Label "\"h\"") (Param [I] None)

          UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam (Param
[V] None) (Param [V] None))
            LeafF (Label "\"g\"") (Param [V] None)

            LeafF (Label "\"h\"") (Param [V] None)

        ConcatenateF [] []
          ConcatenateF [] []
            LeafF (Label "\"g\"") (Param [I] None)
              LeafF (Label "\"g\"") (Param [I] None)

            LeafF (Label "\"h\"") (Param [IV] None)
              LeafF (Label "\"h\"") (Param [IV] None)

          ConcatenateF [] []
            LeafF (Label "\"i\"") (Param [V] None)
              LeafF (Label "\"i\"") (Param [V] None)

            LeafF (Label "\"j\"") (Param [I] Strong)
              LeafF (Label "\"j\"") (Param [I] Strong)

    SimpleBinaryF [] []
      HybridAntContF {antecedentF = [], continuationF = []}
        SentenceF {basicIdeaF = Label "\"a\"", presentationF = [], continuationF =
[]}
          SeqF (LocalVar 0) [Param [I] None,Param [V] None]
            UseIdeaWithF (LocalVar 0) (Param [I] None)
              LeafF (Label "\"a\"") (Param [I] None)

            UseIdeaWithF (LocalVar 0) (Param [V] None)
              LeafF (Label "\"a\"") (Param [V] None)

          ConcatenateF [Label "\"a\""] [Label "\"a\""]
            LeafF (LocalVar 0) (Param [I] None)
              LeafF (Label "\"a\"") (Param [I] None)

            LeafF (Label "\"b\"") (Param [V] Weak)
              LeafF (Label "\"b\"") (Param [V] Weak)

        ConcatenateF [] []
          TwiceFormalUnitF []
            LeafF (Label "\"c\"") (Param [I] None)
              LeafF (Label "\"c\"") (Param [I] None)

            LeafF (Label "\"c\"") (Param [I] None)
              LeafF (Label "\"c\"") (Param [I] None)

          ConcatenateF [] []
            LeafF (Label "\"d\"") (Param [I] None)
```

```
              LeafF (Label "\"d\"") (Param [I] None)

            LeafF (Label "\"e\"") (Param [I] Strong)
              LeafF (Label "\"e\"") (Param [I] Strong)

      HybridAntContF {antecedentF = [], continuationF = []}
        SentenceF {basicIdeaF = Label "\"a\"", presentationF = [], continuationF =
[]}
          SeqF (LocalVar 0) [Param [I] None,Param [V] None]
            UseIdeaWithF (LocalVar 0) (Param [I] None)
              LeafF (Label "\"a\"") (Param [I] None)

            UseIdeaWithF (LocalVar 0) (Param [V] None)
              LeafF (Label "\"a\"") (Param [V] None)

          ConcatenateF [Label "\"a\""] [Label "\"a\""]
            LeafF (LocalVar 0) (Param [I] None)
              LeafF (Label "\"a\"") (Param [I] None)

            LeafF (Label "\"b\"") (Param [V] Weak)
              LeafF (Label "\"b\"") (Param [V] Weak)

        ConcatenateF [] []
          TwiceFormalUnitF []
            LeafF (Label "\"c\"") (Param [I] None)
              LeafF (Label "\"c\"") (Param [I] None)

            LeafF (Label "\"c\"") (Param [I] None)
              LeafF (Label "\"c\"") (Param [I] None)

          ConcatenateF [] []
            LeafF (Label "\"d\"") (Param [I] None)
              LeafF (Label "\"d\"") (Param [I] None)

            LeafF (Label "\"e\"") (Param [I] Strong)
              LeafF (Label "\"e\"") (Param [I] Strong)

      HybridAntContF {antecedentF = [], continuationF = []}
        ConcatenateF [] []
          SeqF (Label "\"a\"") [Param [I] None,Param [V] None]
            UseIdeaWithF (Label "\"a\"") (Param [I] None)
              LeafF (Label "\"a\"") (Param [I] None)

            UseIdeaWithF (Label "\"a\"") (Param [V] None)
              LeafF (Label "\"a\"") (Param [V] None)

          ConcatenateF [] []
            LeafF (Label "\"f\"") (Param [V] None)
              LeafF (Label "\"f\"") (Param [V] None)

            LeafF (Label "\"b\"") (Param [V] Weak)
              LeafF (Label "\"b\"") (Param [V] Weak)

        ConcatenateF [] []
          TwiceFormalUnitF []
            LeafF (Label "\"c\"") (Param [I] None)
```

```
              LeafF (Label "\"c\"") (Param [I] None)

            LeafF (Label "\"c\"") (Param [I] None)
              LeafF (Label "\"c\"") (Param [I] None)

          ConcatenateF [] []
            LeafF (Label "\"d\"") (Param [V] None)
              LeafF (Label "\"d\"") (Param [V] None)

            LeafF (Label "\"e\"") (Param [I] Strong)
              LeafF (Label "\"e\"") (Param [I] Strong)

      HybridAntContF {antecedentF = [], continuationF = []}
        ConcatenateF [] []
          SeqF (Label "\"a\"") [Param [I] None,Param [V] None]
            UseIdeaWithF (Label "\"a\"") (Param [I] None)
              LeafF (Label "\"a\"") (Param [I] None)

            UseIdeaWithF (Label "\"a\"") (Param [V] None)
              LeafF (Label "\"a\"") (Param [V] None)

          ConcatenateF [] []
            LeafF (Label "\"f\"") (Param [V] None)
              LeafF (Label "\"f\"") (Param [V] None)

            LeafF (Label "\"b\"") (Param [V] Weak)
              LeafF (Label "\"b\"") (Param [V] Weak)

        ConcatenateF [] []
          TwiceFormalUnitF []
            LeafF (Label "\"c\"") (Param [I] None)
              LeafF (Label "\"c\"") (Param [I] None)

            LeafF (Label "\"c\"") (Param [I] None)
              LeafF (Label "\"c\"") (Param [I] None)

          ConcatenateF [] []
            LeafF (Label "\"d\"") (Param [V] None)
              LeafF (Label "\"d\"") (Param [V] None)

            LeafF (Label "\"e\"") (Param [I] Strong)
              LeafF (Label "\"e\"") (Param [I] Strong)




  ConcatenateF [] []
    TwiceFormalUnitF []
      HybridAntContF {antecedentF = [], continuationF = []}
        SentenceF {basicIdeaF = Label "\"a\"", presentationF = [], continuationF =
[]}
          SeqF (LocalVar 0) [Param [I] None,Param [V] None]
            UseIdeaWithF (LocalVar 0) (Param [I] None)
              LeafF (Label "\"a\"") (Param [I] None)

            UseIdeaWithF (LocalVar 0) (Param [V] None)
```

```
              LeafF (Label "\"a\"") (Param [V] None)

            ConcatenateF [Label "\"a\""] [Label "\"a\""]
              LeafF (LocalVar 0) (Param [I] None)
                LeafF (Label "\"a\"") (Param [I] None)

              LeafF (Label "\"b\"") (Param [V] Weak)
                LeafF (Label "\"b\"") (Param [V] Weak)

          ConcatenateF [] []
            TwiceFormalUnitF []
              LeafF (Label "\"c\"") (Param [I] None)
                LeafF (Label "\"c\"") (Param [I] None)

              LeafF (Label "\"c\"") (Param [I] None)
                LeafF (Label "\"c\"") (Param [I] None)

            ConcatenateF [] []
              LeafF (Label "\"d\"") (Param [I] None)
                LeafF (Label "\"d\"") (Param [I] None)

              LeafF (Label "\"e\"") (Param [I] Strong)
                LeafF (Label "\"e\"") (Param [I] Strong)

      HybridAntContF {antecedentF = [], continuationF = []}
        SentenceF {basicIdeaF = Label "\"a\"", presentationF = [], continuationF =
[]}
          SeqF (LocalVar 0) [Param [I] None,Param [V] None]
            UseIdeaWithF (LocalVar 0) (Param [I] None)
              LeafF (Label "\"a\"") (Param [I] None)

            UseIdeaWithF (LocalVar 0) (Param [V] None)
              LeafF (Label "\"a\"") (Param [V] None)

            ConcatenateF [Label "\"a\""] [Label "\"a\""]
              LeafF (LocalVar 0) (Param [I] None)
                LeafF (Label "\"a\"") (Param [I] None)

              LeafF (Label "\"b\"") (Param [V] Weak)
                LeafF (Label "\"b\"") (Param [V] Weak)

          ConcatenateF [] []
            TwiceFormalUnitF []
              LeafF (Label "\"c\"") (Param [I] None)
                LeafF (Label "\"c\"") (Param [I] None)

              LeafF (Label "\"c\"") (Param [I] None)
                LeafF (Label "\"c\"") (Param [I] None)

            ConcatenateF [] []
              LeafF (Label "\"d\"") (Param [I] None)
                LeafF (Label "\"d\"") (Param [I] None)

              LeafF (Label "\"e\"") (Param [I] Strong)
                LeafF (Label "\"e\"") (Param [I] Strong)
```

```
TernaryF [] []
  TwiceFormalUnitF []
    HybridAntContF {antecedentF = [], continuationF = []}
      ConcatenateF [] []
        SeqF (Label "\"a\"") [Param [I] None,Param [V] None]
          UseIdeaWithF (Label "\"a\"") (Param [I] None)
            LeafF (Label "\"a\"") (Param [I] None)

          UseIdeaWithF (Label "\"a\"") (Param [V] None)
            LeafF (Label "\"a\"") (Param [V] None)

        ConcatenateF [] []
          LeafF (Label "\"f\"") (Param [V] None)
            LeafF (Label "\"f\"") (Param [V] None)

          LeafF (Label "\"b\"") (Param [V] Weak)
            LeafF (Label "\"b\"") (Param [V] Weak)

      ConcatenateF [] []
        TwiceFormalUnitF []
          LeafF (Label "\"c\"") (Param [I] None)
            LeafF (Label "\"c\"") (Param [I] None)

          LeafF (Label "\"c\"") (Param [I] None)
            LeafF (Label "\"c\"") (Param [I] None)

        ConcatenateF [] []
          LeafF (Label "\"d\"") (Param [V] None)
            LeafF (Label "\"d\"") (Param [V] None)

          LeafF (Label "\"e\"") (Param [I] Strong)
            LeafF (Label "\"e\"") (Param [I] Strong)

    HybridAntContF {antecedentF = [], continuationF = []}
      ConcatenateF [] []
        SeqF (Label "\"a\"") [Param [I] None,Param [V] None]
          UseIdeaWithF (Label "\"a\"") (Param [I] None)
            LeafF (Label "\"a\"") (Param [I] None)

          UseIdeaWithF (Label "\"a\"") (Param [V] None)
            LeafF (Label "\"a\"") (Param [V] None)

        ConcatenateF [] []
          LeafF (Label "\"f\"") (Param [V] None)
            LeafF (Label "\"f\"") (Param [V] None)

          LeafF (Label "\"b\"") (Param [V] Weak)
            LeafF (Label "\"b\"") (Param [V] Weak)

      ConcatenateF [] []
        TwiceFormalUnitF []
          LeafF (Label "\"c\"") (Param [I] None)
            LeafF (Label "\"c\"") (Param [I] None)

          LeafF (Label "\"c\"") (Param [I] None)
            LeafF (Label "\"c\"") (Param [I] None)
```

```
            ConcatenateF [] []
              LeafF (Label "\"d\"") (Param [V] None)
                LeafF (Label "\"d\"") (Param [V] None)

              LeafF (Label "\"e\"") (Param [I] Strong)
                LeafF (Label "\"e\"") (Param [I] Strong)

      ConcatenateF [] []
        RoundedBinaryF {expositionF = [], contrastingMiddleF = []}
          ConcatenateF [] []
            SeqF (Concat (Label "\"g\"") (Label "\"h\"")) [ConcatParam (Param [I]
None) (Param [I] None),ConcatParam (Param [V] None) (Param [V] None)]
              UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam
(Param [I] None) (Param [I] None))
                LeafF (Label "\"g\"") (Param [I] None)

                LeafF (Label "\"h\"") (Param [I] None)

              UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam
(Param [V] None) (Param [V] None))
                LeafF (Label "\"g\"") (Param [V] None)

                LeafF (Label "\"h\"") (Param [V] None)

            ConcatenateF [] []
              ConcatenateF [] []
                LeafF (Label "\"g\"") (Param [I] None)
                  LeafF (Label "\"g\"") (Param [I] None)

                LeafF (Label "\"h\"") (Param [IV] None)
                  LeafF (Label "\"h\"") (Param [IV] None)

              ConcatenateF [] []
                LeafF (Label "\"i\"") (Param [V] None)
                  LeafF (Label "\"i\"") (Param [V] None)

                LeafF (Label "\"j\"") (Param [I] Strong)
                  LeafF (Label "\"j\"") (Param [I] Strong)

          ConcatenateF [] []
            SeqF (Concat (Label "\"g\"") (Label "\"h\"")) [ConcatParam (Param [I]
None) (Param [I] None),ConcatParam (Param [V] None) (Param [V] None)]
              UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam
(Param [I] None) (Param [I] None))
                LeafF (Label "\"g\"") (Param [I] None)

                LeafF (Label "\"h\"") (Param [I] None)

              UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam
(Param [V] None) (Param [V] None))
                LeafF (Label "\"g\"") (Param [V] None)

                LeafF (Label "\"h\"") (Param [V] None)

            ConcatenateF [] []
```

```
              ConcatenateF [] []
                LeafF (Label "\"g\"") (Param [I] None)
                  LeafF (Label "\"g\"") (Param [I] None)

                LeafF (Label "\"h\"") (Param [IV] None)
                  LeafF (Label "\"h\"") (Param [IV] None)

              ConcatenateF [] []
                LeafF (Label "\"i\"") (Param [V] None)
                  LeafF (Label "\"i\"") (Param [V] None)

                LeafF (Label "\"j\"") (Param [I] Strong)
                  LeafF (Label "\"j\"") (Param [I] Strong)

          SeqF (Concat (Label "\"g\"") (Label "\"k\"")) [ConcatParam (Param [I] None)
(Param [IV] None),ConcatParam (Param [II] None) (Param [V] None)]
            UseIdeaWithF (Concat (Label "\"g\"") (Label "\"k\"")) (ConcatParam (Param
[I] None) (Param [IV] None))
              LeafF (Label "\"g\"") (Param [I] None)

              LeafF (Label "\"k\"") (Param [IV] None)

            UseIdeaWithF (Concat (Label "\"g\"") (Label "\"k\"")) (ConcatParam (Param
[II] None) (Param [V] None))
              LeafF (Label "\"g\"") (Param [II] None)

              LeafF (Label "\"k\"") (Param [V] None)

          ConcatenateF [] []
            SeqF (Concat (Label "\"g\"") (Label "\"h\"")) [ConcatParam (Param [I]
None) (Param [I] None),ConcatParam (Param [V] None) (Param [V] None)]
              UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam
(Param [I] None) (Param [I] None))
                LeafF (Label "\"g\"") (Param [I] None)

                LeafF (Label "\"h\"") (Param [I] None)

              UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam
(Param [V] None) (Param [V] None))
                LeafF (Label "\"g\"") (Param [V] None)

                LeafF (Label "\"h\"") (Param [V] None)

            ConcatenateF [] []
              ConcatenateF [] []
                LeafF (Label "\"g\"") (Param [I] None)
                  LeafF (Label "\"g\"") (Param [I] None)

                LeafF (Label "\"h\"") (Param [IV] None)
                  LeafF (Label "\"h\"") (Param [IV] None)

              ConcatenateF [] []
                LeafF (Label "\"i\"") (Param [V] None)
                  LeafF (Label "\"i\"") (Param [V] None)

                LeafF (Label "\"j\"") (Param [I] Strong)
```

```
                LeafF (Label "\"j\"") (Param [I] Strong)

          SeqF (Concat (Label "\"g\"") (Label "\"k\"")) [ConcatParam (Param [I] None)
(Param [IV] None),ConcatParam (Param [II] None) (Param [V] None)]
            UseIdeaWithF (Concat (Label "\"g\"") (Label "\"k\"")) (ConcatParam (Param
[I] None) (Param [IV] None))
              LeafF (Label "\"g\"") (Param [I] None)

              LeafF (Label "\"k\"") (Param [IV] None)

            UseIdeaWithF (Concat (Label "\"g\"") (Label "\"k\"")) (ConcatParam (Param
[II] None) (Param [V] None))
              LeafF (Label "\"g\"") (Param [II] None)

              LeafF (Label "\"k\"") (Param [V] None)

          ConcatenateF [] []
            SeqF (Concat (Label "\"g\"") (Label "\"h\"")) [ConcatParam (Param [I]
None) (Param [I] None),ConcatParam (Param [V] None) (Param [V] None)]
              UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam
(Param [I] None) (Param [I] None))
                LeafF (Label "\"g\"") (Param [I] None)

                LeafF (Label "\"h\"") (Param [I] None)

              UseIdeaWithF (Concat (Label "\"g\"") (Label "\"h\"")) (ConcatParam
(Param [V] None) (Param [V] None))
                LeafF (Label "\"g\"") (Param [V] None)

                LeafF (Label "\"h\"") (Param [V] None)

            ConcatenateF [] []
              ConcatenateF [] []
                LeafF (Label "\"g\"") (Param [I] None)
                  LeafF (Label "\"g\"") (Param [I] None)

                LeafF (Label "\"h\"") (Param [IV] None)
                  LeafF (Label "\"h\"") (Param [IV] None)

              ConcatenateF [] []
                LeafF (Label "\"i\"") (Param [V] None)
                  LeafF (Label "\"i\"") (Param [V] None)

                LeafF (Label "\"j\"") (Param [I] Strong)
                  LeafF (Label "\"j\"") (Param [I] Strong)

        TwiceFormalUnitF []
          HybridAntContF {antecedentF = [], continuationF = []}
            SentenceF {basicIdeaF = Label "\"a\"", presentationF = [], continuationF
= []}
              SeqF (LocalVar 0) [Param [I] None,Param [V] None]
                UseIdeaWithF (LocalVar 0) (Param [I] None)
                  LeafF (Label "\"a\"") (Param [I] None)

                UseIdeaWithF (LocalVar 0) (Param [V] None)
                  LeafF (Label "\"a\"") (Param [V] None)
```

```
              ConcatenateF [Label "\"a\""] [Label "\"a\""]
                LeafF (LocalVar 0) (Param [I] None)
                  LeafF (Label "\"a\"") (Param [I] None)

                LeafF (Label "\"b\"") (Param [V] Weak)
                  LeafF (Label "\"b\"") (Param [V] Weak)

            ConcatenateF [] []
              TwiceFormalUnitF []
                LeafF (Label "\"c\"") (Param [I] None)
                  LeafF (Label "\"c\"") (Param [I] None)

                LeafF (Label "\"c\"") (Param [I] None)
                  LeafF (Label "\"c\"") (Param [I] None)

              ConcatenateF [] []
                LeafF (Label "\"d\"") (Param [I] None)
                  LeafF (Label "\"d\"") (Param [I] None)

                LeafF (Label "\"e\"") (Param [I] Strong)
                  LeafF (Label "\"e\"") (Param [I] Strong)

          HybridAntContF {antecedentF = [], continuationF = []}
            SentenceF {basicIdeaF = Label "\"a\"", presentationF = [], continuationF
= []}
              SeqF (LocalVar 0) [Param [I] None,Param [V] None]
                UseIdeaWithF (LocalVar 0) (Param [I] None)
                  LeafF (Label "\"a\"") (Param [I] None)

                UseIdeaWithF (LocalVar 0) (Param [V] None)
                  LeafF (Label "\"a\"") (Param [V] None)

              ConcatenateF [Label "\"a\""] [Label "\"a\""]
                LeafF (LocalVar 0) (Param [I] None)
                  LeafF (Label "\"a\"") (Param [I] None)

                LeafF (Label "\"b\"") (Param [V] Weak)
                  LeafF (Label "\"b\"") (Param [V] Weak)

            ConcatenateF [] []
              TwiceFormalUnitF []
                LeafF (Label "\"c\"") (Param [I] None)
                  LeafF (Label "\"c\"") (Param [I] None)

                LeafF (Label "\"c\"") (Param [I] None)
                  LeafF (Label "\"c\"") (Param [I] None)

              ConcatenateF [] []
                LeafF (Label "\"d\"") (Param [I] None)
                  LeafF (Label "\"d\"") (Param [I] None)

                LeafF (Label "\"e\"") (Param [I] Strong)
                  LeafF (Label "\"e\"") (Param [I] Strong)

      TwiceFormalUnitF []
```

```
HybridAntContF {antecedentF = [], continuationF = []}
  ConcatenateF [] []
    SeqF (Label "\"a\"") [Param [I] None,Param [V] None]
      UseIdeaWithF (Label "\"a\"") (Param [I] None)
        LeafF (Label "\"a\"") (Param [I] None)

      UseIdeaWithF (Label "\"a\"") (Param [V] None)
        LeafF (Label "\"a\"") (Param [V] None)

    ConcatenateF [] []
      LeafF (Label "\"f\"") (Param [V] None)
        LeafF (Label "\"f\"") (Param [V] None)

      LeafF (Label "\"b\"") (Param [V] Weak)
        LeafF (Label "\"b\"") (Param [V] Weak)

  ConcatenateF [] []
    TwiceFormalUnitF []
      LeafF (Label "\"c\"") (Param [I] None)
        LeafF (Label "\"c\"") (Param [I] None)

      LeafF (Label "\"c\"") (Param [I] None)
        LeafF (Label "\"c\"") (Param [I] None)

    ConcatenateF [] []
      LeafF (Label "\"d\"") (Param [V] None)
        LeafF (Label "\"d\"") (Param [V] None)

      LeafF (Label "\"e\"") (Param [I] Strong)
        LeafF (Label "\"e\"") (Param [I] Strong)

HybridAntContF {antecedentF = [], continuationF = []}
  ConcatenateF [] []
    SeqF (Label "\"a\"") [Param [I] None,Param [V] None]
      UseIdeaWithF (Label "\"a\"") (Param [I] None)
        LeafF (Label "\"a\"") (Param [I] None)

      UseIdeaWithF (Label "\"a\"") (Param [V] None)
        LeafF (Label "\"a\"") (Param [V] None)

    ConcatenateF [] []
      LeafF (Label "\"f\"") (Param [V] None)
        LeafF (Label "\"f\"") (Param [V] None)

      LeafF (Label "\"b\"") (Param [V] Weak)
        LeafF (Label "\"b\"") (Param [V] Weak)

  ConcatenateF [] []
    TwiceFormalUnitF []
      LeafF (Label "\"c\"") (Param [I] None)
        LeafF (Label "\"c\"") (Param [I] None)

      LeafF (Label "\"c\"") (Param [I] None)
        LeafF (Label "\"c\"") (Param [I] None)

    ConcatenateF [] []
```

```
                LeafF (Label "\"d\"") (Param [V] None)
                  LeafF (Label "\"d\"") (Param [V] None)

                LeafF (Label "\"e\"") (Param [I] Strong)
                  LeafF (Label "\"e\"") (Param [I] Strong)

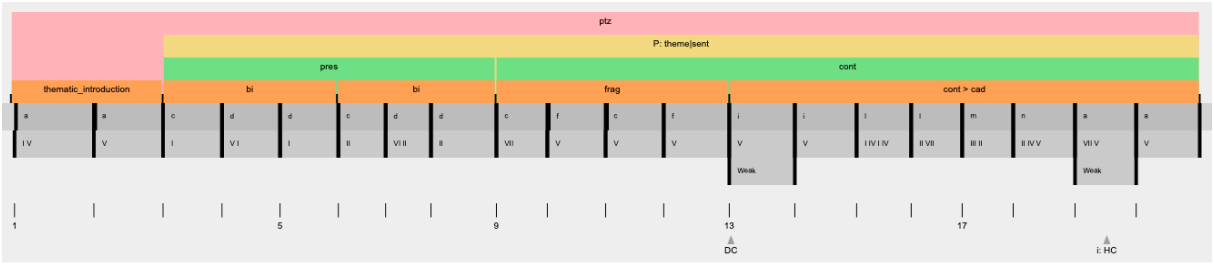
re-generated surface
  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"a\"") (Param [V] None)
  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"b\"") (Param [V] Weak)
  LeafF (Label "\"c\"") (Param [I] None)
  LeafF (Label "\"c\"") (Param [I] None)
  LeafF (Label "\"d\"") (Param [I] None)
  LeafF (Label "\"e\"") (Param [I] Strong)
  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"a\"") (Param [V] None)
  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"b\"") (Param [V] Weak)
  LeafF (Label "\"c\"") (Param [I] None)
  LeafF (Label "\"c\"") (Param [I] None)
  LeafF (Label "\"d\"") (Param [I] None)
  LeafF (Label "\"e\"") (Param [I] Strong)
  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"a\"") (Param [V] None)
  LeafF (Label "\"f\"") (Param [V] None)
  LeafF (Label "\"b\"") (Param [V] Weak)
  LeafF (Label "\"c\"") (Param [I] None)
  LeafF (Label "\"c\"") (Param [I] None)
  LeafF (Label "\"d\"") (Param [V] None)
  LeafF (Label "\"e\"") (Param [I] Strong)
  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"a\"") (Param [V] None)
  LeafF (Label "\"f\"") (Param [V] None)
  LeafF (Label "\"b\"") (Param [V] Weak)
  LeafF (Label "\"c\"") (Param [I] None)
  LeafF (Label "\"c\"") (Param [I] None)
  LeafF (Label "\"d\"") (Param [V] None)
  LeafF (Label "\"e\"") (Param [I] Strong)
  LeafF (Label "\"g\"") (Param [I] None)
  LeafF (Label "\"h\"") (Param [I] None)
  LeafF (Label "\"g\"") (Param [V] None)
  LeafF (Label "\"h\"") (Param [V] None)
  LeafF (Label "\"g\"") (Param [I] None)
  LeafF (Label "\"h\"") (Param [IV] None)
  LeafF (Label "\"i\"") (Param [V] None)
  LeafF (Label "\"j\"") (Param [I] Strong)
  LeafF (Label "\"g\"") (Param [I] None)
  LeafF (Label "\"h\"") (Param [I] None)
  LeafF (Label "\"g\"") (Param [V] None)
  LeafF (Label "\"h\"") (Param [V] None)
  LeafF (Label "\"g\"") (Param [I] None)
  LeafF (Label "\"h\"") (Param [IV] None)
  LeafF (Label "\"i\"") (Param [V] None)
  LeafF (Label "\"j\"") (Param [I] Strong)
  LeafF (Label "\"g\"") (Param [I] None)
```

```
LeafF (Label "\"k\"") (Param [IV] None)
LeafF (Label "\"g\"") (Param [II] None)
LeafF (Label "\"k\"") (Param [V] None)
LeafF (Label "\"g\"") (Param [I] None)
LeafF (Label "\"h\"") (Param [I] None)
LeafF (Label "\"g\"") (Param [V] None)
LeafF (Label "\"h\"") (Param [V] None)
LeafF (Label "\"g\"") (Param [I] None)
LeafF (Label "\"h\"") (Param [IV] None)
LeafF (Label "\"i\"") (Param [V] None)
LeafF (Label "\"j\"") (Param [I] Strong)
LeafF (Label "\"g\"") (Param [I] None)
LeafF (Label "\"k\"") (Param [IV] None)
LeafF (Label "\"g\"") (Param [II] None)
LeafF (Label "\"k\"") (Param [V] None)
LeafF (Label "\"g\"") (Param [I] None)
LeafF (Label "\"h\"") (Param [I] None)
LeafF (Label "\"g\"") (Param [V] None)
LeafF (Label "\"h\"") (Param [V] None)
LeafF (Label "\"g\"") (Param [I] None)
LeafF (Label "\"h\"") (Param [IV] None)
LeafF (Label "\"i\"") (Param [V] None)
LeafF (Label "\"j\"") (Param [I] Strong)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"a\"") (Param [V] None)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"b\"") (Param [V] Weak)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"d\"") (Param [I] None)
LeafF (Label "\"e\"") (Param [I] Strong)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"a\"") (Param [V] None)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"b\"") (Param [V] Weak)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"d\"") (Param [I] None)
LeafF (Label "\"e\"") (Param [I] Strong)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"a\"") (Param [V] None)
LeafF (Label "\"f\"") (Param [V] None)
LeafF (Label "\"b\"") (Param [V] Weak)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"d\"") (Param [V] None)
LeafF (Label "\"e\"") (Param [I] Strong)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"a\"") (Param [V] None)
LeafF (Label "\"f\"") (Param [V] None)
LeafF (Label "\"b\"") (Param [V] Weak)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"d\"") (Param [V] None)
LeafF (Label "\"e\"") (Param [I] Strong)
```

```
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"a\"") (Param [V] None)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"b\"") (Param [V] Weak)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"d\"") (Param [I] None)
LeafF (Label "\"e\"") (Param [I] Strong)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"a\"") (Param [V] None)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"b\"") (Param [V] Weak)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"d\"") (Param [I] None)
LeafF (Label "\"e\"") (Param [I] Strong)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"a\"") (Param [V] None)
LeafF (Label "\"f\"") (Param [V] None)
LeafF (Label "\"b\"") (Param [V] Weak)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"d\"") (Param [V] None)
LeafF (Label "\"e\"") (Param [I] Strong)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"a\"") (Param [V] None)
LeafF (Label "\"f\"") (Param [V] None)
LeafF (Label "\"b\"") (Param [V] Weak)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"d\"") (Param [V] None)
LeafF (Label "\"e\"") (Param [I] Strong)
LeafF (Label "\"g\"") (Param [I] None)
LeafF (Label "\"h\"") (Param [I] None)
LeafF (Label "\"g\"") (Param [V] None)
LeafF (Label "\"h\"") (Param [V] None)
LeafF (Label "\"g\"") (Param [I] None)
LeafF (Label "\"h\"") (Param [IV] None)
LeafF (Label "\"i\"") (Param [V] None)
LeafF (Label "\"j\"") (Param [I] Strong)
LeafF (Label "\"g\"") (Param [I] None)
LeafF (Label "\"h\"") (Param [I] None)
LeafF (Label "\"g\"") (Param [V] None)
LeafF (Label "\"h\"") (Param [V] None)
LeafF (Label "\"g\"") (Param [I] None)
LeafF (Label "\"h\"") (Param [IV] None)
LeafF (Label "\"i\"") (Param [V] None)
LeafF (Label "\"j\"") (Param [I] Strong)
LeafF (Label "\"g\"") (Param [I] None)
LeafF (Label "\"k\"") (Param [IV] None)
LeafF (Label "\"g\"") (Param [II] None)
LeafF (Label "\"k\"") (Param [V] None)
LeafF (Label "\"g\"") (Param [I] None)
LeafF (Label "\"h\"") (Param [I] None)
```

```
LeafF (Label "\"g\"") (Param [V] None)
LeafF (Label "\"h\"") (Param [V] None)
LeafF (Label "\"g\"") (Param [I] None)
LeafF (Label "\"h\"") (Param [IV] None)
LeafF (Label "\"i\"") (Param [V] None)
LeafF (Label "\"j\"") (Param [I] Strong)
LeafF (Label "\"g\"") (Param [I] None)
LeafF (Label "\"k\"") (Param [IV] None)
LeafF (Label "\"g\"") (Param [II] None)
LeafF (Label "\"k\"") (Param [V] None)
LeafF (Label "\"g\"") (Param [I] None)
LeafF (Label "\"h\"") (Param [I] None)
LeafF (Label "\"g\"") (Param [V] None)
LeafF (Label "\"h\"") (Param [V] None)
LeafF (Label "\"g\"") (Param [I] None)
LeafF (Label "\"h\"") (Param [IV] None)
LeafF (Label "\"i\"") (Param [V] None)
LeafF (Label "\"j\"") (Param [I] Strong)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"a\"") (Param [V] None)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"b\"") (Param [V] Weak)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"d\"") (Param [I] None)
LeafF (Label "\"e\"") (Param [I] Strong)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"a\"") (Param [V] None)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"b\"") (Param [V] Weak)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"d\"") (Param [I] None)
LeafF (Label "\"e\"") (Param [I] Strong)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"a\"") (Param [V] None)
LeafF (Label "\"f\"") (Param [V] None)
LeafF (Label "\"b\"") (Param [V] Weak)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"d\"") (Param [V] None)
LeafF (Label "\"e\"") (Param [I] Strong)
LeafF (Label "\"a\"") (Param [I] None)
LeafF (Label "\"a\"") (Param [V] None)
LeafF (Label "\"f\"") (Param [V] None)
LeafF (Label "\"b\"") (Param [V] Weak)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"d\"") (Param [V] None)
LeafF (Label "\"e\"") (Param [I] Strong)
```

Examples from linz annotation (Felipe)
BTHVN_08.txt




```
Input surface:
size = 130
  {a ⟦I-V None⟧}
  {a ⟦V None⟧}
  {c ⟦I None⟧}
  {d ⟦V-I None⟧}
  {d ⟦I None⟧}
  {c ⟦II None⟧}
  {d ⟦VI-II None⟧}
  {d ⟦II None⟧}
  {c ⟦VII None⟧}
  {f ⟦V None⟧}
  {c ⟦V None⟧}
  {f ⟦V None⟧}
  {i ⟦V Weak⟧}
  {i ⟦V None⟧}
  {l ⟦I-IV-I-IV None⟧}
  {l ⟦II-VII None⟧}
  {m ⟦III-II None⟧}
  {n ⟦II-IV-V None⟧}
  {a ⟦VII-V Weak⟧}
  {a ⟦V None⟧}



Output analysis:
  size=123 compressionRate=1.056910569105691
  Concatenate
    Seq
      idea = "a"
      parameters = ⟦I-V None⟧
                   ⟦V None⟧
    Concatenate
      Concatenate
        Seq
          idea = "c"+Twice "d"
          parameters = ConcatParam
                         ⟦I None⟧
                         ConcatParam
                           ⟦V-I None⟧
                           ⟦I None⟧
                       ConcatParam
                         ⟦II None⟧
```

```
                      ConcatParam
                        ⟦VI-II None⟧
                        ⟦II None⟧
      Concatenate
        Seq
          idea = "c"+"f"
          parameters = ConcatParam
                         ⟦VII None⟧
                         ⟦V None⟧
                       ConcatParam
                         ⟦V None⟧
                         ⟦V None⟧
        {"i" ⟦V Weak⟧}
    Concatenate
      {"i" ⟦V None⟧}
      Concatenate
        Seq
          idea = "l"
          parameters = ⟦I-IV-I-IV None⟧
                       ⟦II-VII None⟧
        Concatenate
          {"m" ⟦III-II None⟧}
          Concatenate
            {"n" ⟦II-IV-V None⟧}
            Seq
              idea = "a"
              parameters = ⟦VII-V Weak⟧
                           ⟦V None⟧



size=123 compressionRate=1.056910569105691
Concatenate
  Concatenate
    Seq
      idea = "a"
      parameters = ⟦I-V None⟧
                   ⟦V None⟧
    Seq
      idea = "c"+Twice "d"
      parameters = ConcatParam
                     ⟦I None⟧
                     ConcatParam
                       ⟦V-I None⟧
                       ⟦I None⟧
                   ConcatParam
                     ⟦II None⟧
                     ConcatParam
                       ⟦VI-II None⟧
                       ⟦II None⟧
  Concatenate
    Seq
      idea = "c"+"f"
      parameters = ConcatParam
                     ⟦VII None⟧
                     ⟦V None⟧
```

```
                    ConcatParam
                      ⟦V None⟧
                      ⟦V None⟧
      Concatenate
        {"i" ⟦V Weak⟧}
        Concatenate
          {"i" ⟦V None⟧}
          Concatenate
            Seq
              idea = "l"
              parameters = ⟦I-IV-I-IV None⟧
                           ⟦II-VII None⟧
            Concatenate
              {"m" ⟦III-II None⟧}
              Concatenate
                {"n" ⟦II-IV-V None⟧}
                Seq
                  idea = "a"
                  parameters = ⟦VII-V Weak⟧
                               ⟦V None⟧
expanded analysis:
  ConcatenateF [] []
    SeqF (Label "\"a\"") [Param [I,V] None,Param [V] None]
      UseIdeaWithF (Label "\"a\"") (Param [I,V] None)
        LeafF (Label "\"a\"") (Param [I,V] None)

      UseIdeaWithF (Label "\"a\"") (Param [V] None)
        LeafF (Label "\"a\"") (Param [V] None)

    ConcatenateF [] []
      ConcatenateF [] []
        SeqF (Concat (Label "\"c\"") (Twice (Label "\"d\""))) [ConcatParam (Param [I]
None) (ConcatParam (Param [V,I] None) (Param [I] None)),ConcatParam (Param [II] None)
(ConcatParam (Param [VI,II] None) (Param [II] None))]
          UseIdeaWithF (Concat (Label "\"c\"") (Twice (Label "\"d\""))) (ConcatParam
(Param [I] None) (ConcatParam (Param [V,I] None) (Param [I] None)))
            LeafF (Label "\"c\"") (Param [I] None)

            LeafF (Label "\"d\"") (Param [V,I] None)

            LeafF (Label "\"d\"") (Param [I] None)

          UseIdeaWithF (Concat (Label "\"c\"") (Twice (Label "\"d\""))) (ConcatParam
(Param [II] None) (ConcatParam (Param [VI,II] None) (Param [II] None)))
            LeafF (Label "\"c\"") (Param [II] None)

            LeafF (Label "\"d\"") (Param [VI,II] None)

            LeafF (Label "\"d\"") (Param [II] None)

        ConcatenateF [] []
          SeqF (Concat (Label "\"c\"") (Label "\"f\"")) [ConcatParam (Param [VII]
None) (Param [V] None),ConcatParam (Param [V] None) (Param [V] None)]
            UseIdeaWithF (Concat (Label "\"c\"") (Label "\"f\"")) (ConcatParam (Param
[VII] None) (Param [V] None))
              LeafF (Label "\"c\"") (Param [VII] None)
```

```
              LeafF (Label "\"f\"") (Param [V] None)

            UseIdeaWithF (Concat (Label "\"c\"") (Label "\"f\"")) (ConcatParam (Param
[V] None) (Param [V] None))
              LeafF (Label "\"c\"") (Param [V] None)

              LeafF (Label "\"f\"") (Param [V] None)

          LeafF (Label "\"i\"") (Param [V] Weak)
            LeafF (Label "\"i\"") (Param [V] Weak)

      ConcatenateF [] []
        LeafF (Label "\"i\"") (Param [V] None)
          LeafF (Label "\"i\"") (Param [V] None)

        ConcatenateF [] []
          SeqF (Label "\"l\"") [Param [I,IV,I,IV] None,Param [II,VII] None]
            UseIdeaWithF (Label "\"l\"") (Param [I,IV,I,IV] None)
              LeafF (Label "\"l\"") (Param [I,IV,I,IV] None)

            UseIdeaWithF (Label "\"l\"") (Param [II,VII] None)
              LeafF (Label "\"l\"") (Param [II,VII] None)

          ConcatenateF [] []
            LeafF (Label "\"m\"") (Param [III,II] None)
              LeafF (Label "\"m\"") (Param [III,II] None)

            ConcatenateF [] []
              LeafF (Label "\"n\"") (Param [II,IV,V] None)
                LeafF (Label "\"n\"") (Param [II,IV,V] None)

              SeqF (Label "\"a\"") [Param [VII,V] Weak,Param [V] None]
                UseIdeaWithF (Label "\"a\"") (Param [VII,V] Weak)
                  LeafF (Label "\"a\"") (Param [VII,V] Weak)

                UseIdeaWithF (Label "\"a\"") (Param [V] None)
                  LeafF (Label "\"a\"") (Param [V] None)




  ConcatenateF [] []
    ConcatenateF [] []
      SeqF (Label "\"a\"") [Param [I,V] None,Param [V] None]
        UseIdeaWithF (Label "\"a\"") (Param [I,V] None)
          LeafF (Label "\"a\"") (Param [I,V] None)

        UseIdeaWithF (Label "\"a\"") (Param [V] None)
          LeafF (Label "\"a\"") (Param [V] None)

      SeqF (Concat (Label "\"c\"") (Twice (Label "\"d\""))) [ConcatParam (Param [I]
None) (ConcatParam (Param [V,I] None) (Param [I] None)),ConcatParam (Param [II] None)
(ConcatParam (Param [VI,II] None) (Param [II] None))]
        UseIdeaWithF (Concat (Label "\"c\"") (Twice (Label "\"d\""))) (ConcatParam
(Param [I] None) (ConcatParam (Param [V,I] None) (Param [I] None)))
```

```
          LeafF (Label "\"c\"") (Param [I] None)

          LeafF (Label "\"d\"") (Param [V,I] None)

          LeafF (Label "\"d\"") (Param [I] None)

        UseIdeaWithF (Concat (Label "\"c\"") (Twice (Label "\"d\""))) (ConcatParam
(Param [II] None) (ConcatParam (Param [VI,II] None) (Param [II] None)))
          LeafF (Label "\"c\"") (Param [II] None)

          LeafF (Label "\"d\"") (Param [VI,II] None)

          LeafF (Label "\"d\"") (Param [II] None)

    ConcatenateF [] []
      SeqF (Concat (Label "\"c\"") (Label "\"f\"")) [ConcatParam (Param [VII] None)
(Param [V] None),ConcatParam (Param [V] None) (Param [V] None)]
        UseIdeaWithF (Concat (Label "\"c\"") (Label "\"f\"")) (ConcatParam (Param
[VII] None) (Param [V] None))
          LeafF (Label "\"c\"") (Param [VII] None)

          LeafF (Label "\"f\"") (Param [V] None)

        UseIdeaWithF (Concat (Label "\"c\"") (Label "\"f\"")) (ConcatParam (Param [V]
None) (Param [V] None))
          LeafF (Label "\"c\"") (Param [V] None)

          LeafF (Label "\"f\"") (Param [V] None)

      ConcatenateF [] []
        LeafF (Label "\"i\"") (Param [V] Weak)
          LeafF (Label "\"i\"") (Param [V] Weak)

        ConcatenateF [] []
          LeafF (Label "\"i\"") (Param [V] None)
            LeafF (Label "\"i\"") (Param [V] None)

          ConcatenateF [] []
            SeqF (Label "\"l\"") [Param [I,IV,I,IV] None,Param [II,VII] None]
              UseIdeaWithF (Label "\"l\"") (Param [I,IV,I,IV] None)
                LeafF (Label "\"l\"") (Param [I,IV,I,IV] None)

              UseIdeaWithF (Label "\"l\"") (Param [II,VII] None)
                LeafF (Label "\"l\"") (Param [II,VII] None)

            ConcatenateF [] []
              LeafF (Label "\"m\"") (Param [III,II] None)
                LeafF (Label "\"m\"") (Param [III,II] None)

              ConcatenateF [] []
                LeafF (Label "\"n\"") (Param [II,IV,V] None)
                  LeafF (Label "\"n\"") (Param [II,IV,V] None)

                SeqF (Label "\"a\"") [Param [VII,V] Weak,Param [V] None]
                  UseIdeaWithF (Label "\"a\"") (Param [VII,V] Weak)
                    LeafF (Label "\"a\"") (Param [VII,V] Weak)
```

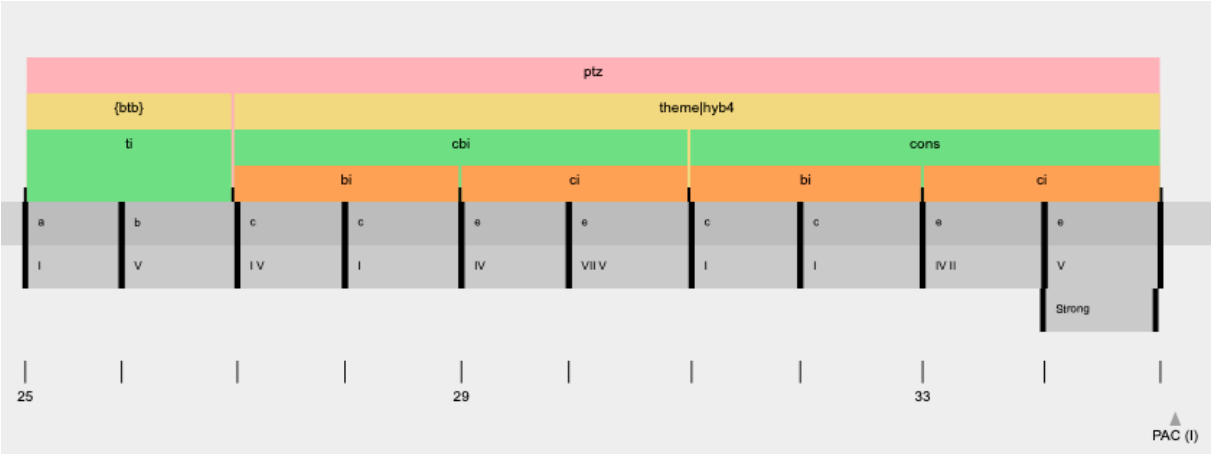
```
                  UseIdeaWithF (Label "\"a\"") (Param [V] None)
                    LeafF (Label "\"a\"") (Param [V] None)

re-generated surface
  LeafF (Label "\"a\"") (Param [I,V] None)
  LeafF (Label "\"a\"") (Param [V] None)
  LeafF (Label "\"c\"") (Param [I] None)
  LeafF (Label "\"d\"") (Param [V,I] None)
  LeafF (Label "\"d\"") (Param [I] None)
  LeafF (Label "\"c\"") (Param [II] None)
  LeafF (Label "\"d\"") (Param [VI,II] None)
  LeafF (Label "\"d\"") (Param [II] None)
  LeafF (Label "\"c\"") (Param [VII] None)
  LeafF (Label "\"f\"") (Param [V] None)
  LeafF (Label "\"c\"") (Param [V] None)
  LeafF (Label "\"f\"") (Param [V] None)
  LeafF (Label "\"i\"") (Param [V] Weak)
  LeafF (Label "\"i\"") (Param [V] None)
  LeafF (Label "\"l\"") (Param [I,IV,I,IV] None)
  LeafF (Label "\"l\"") (Param [II,VII] None)
  LeafF (Label "\"m\"") (Param [III,II] None)
  LeafF (Label "\"n\"") (Param [II,IV,V] None)
  LeafF (Label "\"a\"") (Param [VII,V] Weak)
  LeafF (Label "\"a\"") (Param [V] None)



  LeafF (Label "\"a\"") (Param [I,V] None)
  LeafF (Label "\"a\"") (Param [V] None)
  LeafF (Label "\"c\"") (Param [I] None)
  LeafF (Label "\"d\"") (Param [V,I] None)
  LeafF (Label "\"d\"") (Param [I] None)
  LeafF (Label "\"c\"") (Param [II] None)
  LeafF (Label "\"d\"") (Param [VI,II] None)
  LeafF (Label "\"d\"") (Param [II] None)
  LeafF (Label "\"c\"") (Param [VII] None)
  LeafF (Label "\"f\"") (Param [V] None)
  LeafF (Label "\"c\"") (Param [V] None)
  LeafF (Label "\"f\"") (Param [V] None)
  LeafF (Label "\"i\"") (Param [V] Weak)
  LeafF (Label "\"i\"") (Param [V] None)
  LeafF (Label "\"l\"") (Param [I,IV,I,IV] None)
  LeafF (Label "\"l\"") (Param [II,VII] None)
  LeafF (Label "\"m\"") (Param [III,II] None)
  LeafF (Label "\"n\"") (Param [II,IV,V] None)
  LeafF (Label "\"a\"") (Param [VII,V] Weak)
  LeafF (Label "\"a\"") (Param [V] None)
```

BTHVN_10.txt




```
Input surface:
size = 62
  {a ⟦I None⟧}
  {b ⟦V None⟧}
  {c ⟦I-V None⟧}
  {c ⟦I None⟧}
  {e ⟦IV None⟧}
  {e ⟦VII-V None⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {e ⟦IV-II None⟧}
  {e ⟦V Strong⟧}
```

```
Output analysis:
  size=54 compressionRate=1.1481481481481481
  Concatenate
    {"a" ⟦I None⟧}
    Concatenate
      {"b" ⟦V None⟧}
      Seq
        idea = Twice "c"+Twice "e"
        parameters = ConcatParam
                       ConcatParam
                         ⟦I-V None⟧
                         ⟦I None⟧
                       ConcatParam
                         ⟦IV None⟧
                         ⟦VII-V None⟧
                     ConcatParam
                       ⟦I None⟧
                       ConcatParam
                         ⟦I None⟧
                         ConcatParam
                           ⟦IV-II None⟧
                           ⟦V Strong⟧
```

```
  size=54 compressionRate=1.1481481481481481
  Concatenate
    Concatenate
      {"a" ⟦I None⟧}
      {"b" ⟦V None⟧}
    Seq
      idea = Twice "c"+Twice "e"
      parameters = ConcatParam
                     ConcatParam
                       ⟦I-V None⟧
                       ⟦I None⟧
                     ConcatParam
                       ⟦IV None⟧
                       ⟦VII-V None⟧
                   ConcatParam
                     ⟦I None⟧
                     ConcatParam
                       ⟦I None⟧
                       ConcatParam
                         ⟦IV-II None⟧
                         ⟦V Strong⟧
expanded analysis:
  ConcatenateF [] []
    LeafF (Label "\"a\"") (Param [I] None)
      LeafF (Label "\"a\"") (Param [I] None)

    ConcatenateF [] []
      LeafF (Label "\"b\"") (Param [V] None)
        LeafF (Label "\"b\"") (Param [V] None)

      SeqF (Concat (Twice (Label "\"c\"")) (Twice (Label "\"e\""))) [ConcatParam
(ConcatParam (Param [I,V] None) (Param [I] None)) (ConcatParam (Param [IV] None)
(Param [VII,V] None)),ConcatParam (Param [I] None) (ConcatParam (Param [I] None)
(ConcatParam (Param [IV,II] None) (Param [V] Strong)))]
        UseIdeaWithF (Concat (Twice (Label "\"c\"")) (Twice (Label "\"e\"")))
(ConcatParam (ConcatParam (Param [I,V] None) (Param [I] None)) (ConcatParam (Param
[IV] None) (Param [VII,V] None)))
          LeafF (Label "\"c\"") (Param [I,V] None)

          LeafF (Label "\"c\"") (Param [I] None)

          LeafF (Label "\"e\"") (Param [IV] None)

          LeafF (Label "\"e\"") (Param [VII,V] None)

        UseIdeaWithF (Concat (Twice (Label "\"c\"")) (Twice (Label "\"e\"")))
(ConcatParam (Param [I] None) (ConcatParam (Param [I] None) (ConcatParam (Param
[IV,II] None) (Param [V] Strong))))
          LeafF (Label "\"c\"") (Param [I] None)

          LeafF (Label "\"c\"") (Param [I] None)

          LeafF (Label "\"e\"") (Param [IV,II] None)

          LeafF (Label "\"e\"") (Param [V] Strong)
```

```
  ConcatenateF [] []
    ConcatenateF [] []
      LeafF (Label "\"a\"") (Param [I] None)
        LeafF (Label "\"a\"") (Param [I] None)

      LeafF (Label "\"b\"") (Param [V] None)
        LeafF (Label "\"b\"") (Param [V] None)

    SeqF (Concat (Twice (Label "\"c\"")) (Twice (Label "\"e\""))) [ConcatParam
(ConcatParam (Param [I,V] None) (Param [I] None)) (ConcatParam (Param [IV] None)
(Param [VII,V] None)),ConcatParam (Param [I] None) (ConcatParam (Param [I] None)
(ConcatParam (Param [IV,II] None) (Param [V] Strong)))]
      UseIdeaWithF (Concat (Twice (Label "\"c\"")) (Twice (Label "\"e\"")))
(ConcatParam (ConcatParam (Param [I,V] None) (Param [I] None)) (ConcatParam (Param
[IV] None) (Param [VII,V] None)))
        LeafF (Label "\"c\"") (Param [I,V] None)

        LeafF (Label "\"c\"") (Param [I] None)

        LeafF (Label "\"e\"") (Param [IV] None)

        LeafF (Label "\"e\"") (Param [VII,V] None)

      UseIdeaWithF (Concat (Twice (Label "\"c\"")) (Twice (Label "\"e\"")))
(ConcatParam (Param [I] None) (ConcatParam (Param [I] None) (ConcatParam (Param
[IV,II] None) (Param [V] Strong))))
        LeafF (Label "\"c\"") (Param [I] None)

        LeafF (Label "\"c\"") (Param [I] None)

        LeafF (Label "\"e\"") (Param [IV,II] None)

        LeafF (Label "\"e\"") (Param [V] Strong)

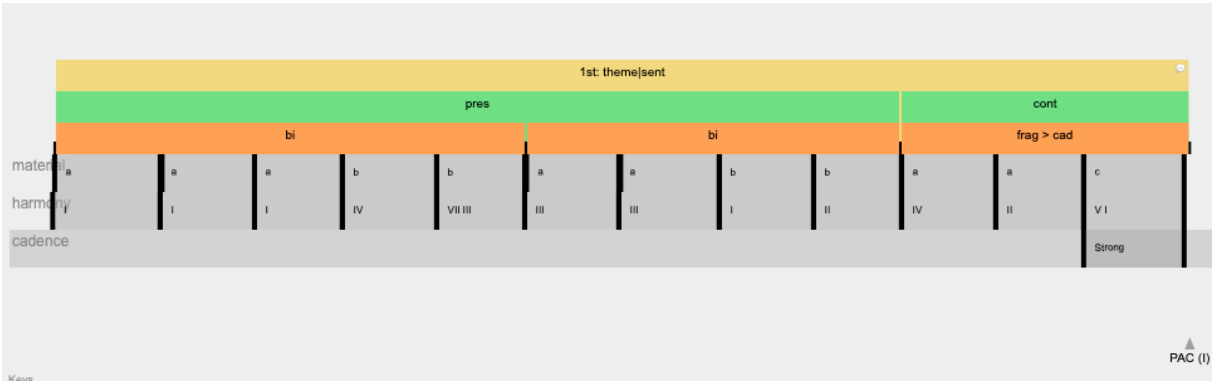
re-generated surface
  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"b\"") (Param [V] None)
  LeafF (Label "\"c\"") (Param [I,V] None)
  LeafF (Label "\"c\"") (Param [I] None)
  LeafF (Label "\"e\"") (Param [IV] None)
  LeafF (Label "\"e\"") (Param [VII,V] None)
  LeafF (Label "\"c\"") (Param [I] None)
  LeafF (Label "\"c\"") (Param [I] None)
  LeafF (Label "\"e\"") (Param [IV,II] None)
  LeafF (Label "\"e\"") (Param [V] Strong)



  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"b\"") (Param [V] None)
  LeafF (Label "\"c\"") (Param [I,V] None)
  LeafF (Label "\"c\"") (Param [I] None)
```

```
LeafF (Label "\"e\"") (Param [IV] None)
LeafF (Label "\"e\"") (Param [VII,V] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"c\"") (Param [I] None)
LeafF (Label "\"e\"") (Param [IV,II] None)
LeafF (Label "\"e\"") (Param [V] Strong)
```

MNDLSSHN_3.txt




```
Input surface:
size = 73
  {a ⟦I None⟧}
  {a ⟦I None⟧}
  {a ⟦I None⟧}
  {b ⟦IV None⟧}
  {b ⟦VII-III None⟧}
  {a ⟦III None⟧}
  {a ⟦III None⟧}
  {b ⟦I None⟧}
  {b ⟦II None⟧}
  {a ⟦IV None⟧}
  {a ⟦II None⟧}
  {c ⟦V-I Strong⟧}



Output analysis:
  size=61 compressionRate=1.1967213114754098
  Sentence
    basicIdea = "a"
    presentation = TwiceFormalUnit
                     {$0 ⟦I None⟧}
    continuation = Concatenate
                     {$0 ⟦I None⟧}
                     Concatenate
                       Seq
                         idea = Twice "b"+Twice $0
                         parameters = ConcatParam
                                        ConcatParam
                                          ⟦IV None⟧
                                          ⟦VII-III None⟧
                                        ConcatParam
                                          ⟦III None⟧
                                          ⟦III None⟧
                                      ConcatParam
                                        ConcatParam
                                          ⟦I None⟧
                                          ⟦II None⟧
                                        ConcatParam
                                          ⟦IV None⟧
```

```
                                        ⟦II None⟧
                      {"c" ⟦V-I Strong⟧}



  size=61 compressionRate=1.1967213114754098
  Concatenate
    {"a" ⟦I None⟧}
    Sentence
      basicIdea = "a"
      presentation = TwiceFormalUnit
                      {$0 ⟦I None⟧}
      continuation = Concatenate
                      Seq
                        idea = Twice "b"+Twice $0
                        parameters = ConcatParam
                                       ConcatParam
                                         ⟦IV None⟧
                                         ⟦VII-III None⟧
                                       ConcatParam
                                         ⟦III None⟧
                                         ⟦III None⟧
                                     ConcatParam
                                       ConcatParam
                                         ⟦I None⟧
                                         ⟦II None⟧
                                       ConcatParam
                                         ⟦IV None⟧
                                         ⟦II None⟧
                      {"c" ⟦V-I Strong⟧}
expanded analysis:
  SentenceF {basicIdeaF = Label "\"a\"", presentationF = [], continuationF = []}
    TwiceFormalUnitF [Label "\"a\""]
      LeafF (LocalVar 0) (Param [I] None)
        LeafF (Label "\"a\"") (Param [I] None)

      LeafF (LocalVar 0) (Param [I] None)
        LeafF (Label "\"a\"") (Param [I] None)

    ConcatenateF [Label "\"a\""] [Label "\"a\""]
      LeafF (LocalVar 0) (Param [I] None)
        LeafF (Label "\"a\"") (Param [I] None)

      ConcatenateF [Label "\"a\""] [Label "\"a\""]
        SeqF (Concat (Twice (Label "\"b\"")) (Twice (LocalVar 0))) [ConcatParam
(ConcatParam (Param [IV] None) (Param [VII,III] None)) (ConcatParam (Param [III]
None) (Param [III] None)),ConcatParam (ConcatParam (Param [I] None) (Param [II]
None)) (ConcatParam (Param [IV] None) (Param [II] None))]
          UseIdeaWithF (Concat (Twice (Label "\"b\"")) (Twice (LocalVar 0)))
(ConcatParam (ConcatParam (Param [IV] None) (Param [VII,III] None)) (ConcatParam
(Param [III] None) (Param [III] None)))
            LeafF (Label "\"b\"") (Param [IV] None)

            LeafF (Label "\"b\"") (Param [VII,III] None)

            LeafF (Label "\"a\"") (Param [III] None)
```

```
            LeafF (Label "\"a\"") (Param [III] None)

          UseIdeaWithF (Concat (Twice (Label "\"b\"")) (Twice (LocalVar 0)))
(ConcatParam (ConcatParam (Param [I] None) (Param [II] None)) (ConcatParam (Param
[IV] None) (Param [II] None)))
            LeafF (Label "\"b\"") (Param [I] None)

            LeafF (Label "\"b\"") (Param [II] None)

            LeafF (Label "\"a\"") (Param [IV] None)

            LeafF (Label "\"a\"") (Param [II] None)

        LeafF (Label "\"c\"") (Param [V,I] Strong)
          LeafF (Label "\"c\"") (Param [V,I] Strong)




  ConcatenateF [] []
    LeafF (Label "\"a\"") (Param [I] None)
      LeafF (Label "\"a\"") (Param [I] None)

    SentenceF {basicIdeaF = Label "\"a\"", presentationF = [], continuationF = []}
      TwiceFormalUnitF [Label "\"a\""]
        LeafF (LocalVar 0) (Param [I] None)
          LeafF (Label "\"a\"") (Param [I] None)

        LeafF (LocalVar 0) (Param [I] None)
          LeafF (Label "\"a\"") (Param [I] None)

      ConcatenateF [Label "\"a\""] [Label "\"a\""]
        SeqF (Concat (Twice (Label "\"b\"")) (Twice (LocalVar 0))) [ConcatParam
(ConcatParam (Param [IV] None) (Param [VII,III] None)) (ConcatParam (Param [III]
None) (Param [III] None)),ConcatParam (ConcatParam (Param [I] None) (Param [II]
None)) (ConcatParam (Param [IV] None) (Param [II] None))]
          UseIdeaWithF (Concat (Twice (Label "\"b\"")) (Twice (LocalVar 0)))
(ConcatParam (ConcatParam (Param [IV] None) (Param [VII,III] None)) (ConcatParam
(Param [III] None) (Param [III] None)))
            LeafF (Label "\"b\"") (Param [IV] None)

            LeafF (Label "\"b\"") (Param [VII,III] None)

            LeafF (Label "\"a\"") (Param [III] None)

            LeafF (Label "\"a\"") (Param [III] None)

          UseIdeaWithF (Concat (Twice (Label "\"b\"")) (Twice (LocalVar 0)))
(ConcatParam (ConcatParam (Param [I] None) (Param [II] None)) (ConcatParam (Param
[IV] None) (Param [II] None)))
            LeafF (Label "\"b\"") (Param [I] None)

            LeafF (Label "\"b\"") (Param [II] None)

            LeafF (Label "\"a\"") (Param [IV] None)
```

```
            LeafF (Label "\"a\"") (Param [II] None)

          LeafF (Label "\"c\"") (Param [V,I] Strong)
            LeafF (Label "\"c\"") (Param [V,I] Strong)

re-generated surface
  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"b\"") (Param [IV] None)
  LeafF (Label "\"b\"") (Param [VII,III] None)
  LeafF (Label "\"a\"") (Param [III] None)
  LeafF (Label "\"a\"") (Param [III] None)
  LeafF (Label "\"b\"") (Param [I] None)
  LeafF (Label "\"b\"") (Param [II] None)
  LeafF (Label "\"a\"") (Param [IV] None)
  LeafF (Label "\"a\"") (Param [II] None)
  LeafF (Label "\"c\"") (Param [V,I] Strong)



  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"a\"") (Param [I] None)
  LeafF (Label "\"b\"") (Param [IV] None)
  LeafF (Label "\"b\"") (Param [VII,III] None)
  LeafF (Label "\"a\"") (Param [III] None)
  LeafF (Label "\"a\"") (Param [III] None)
  LeafF (Label "\"b\"") (Param [I] None)
  LeafF (Label "\"b\"") (Param [II] None)
  LeafF (Label "\"a\"") (Param [IV] None)
  LeafF (Label "\"a\"") (Param [II] None)
  LeafF (Label "\"c\"") (Param [V,I] Strong)
```